Asif Ahmad Najar Asif Ahmad Najar

Asif Ahmad Najar - CSG_16_Report.docx

Quick Submit

Quick Submit

Presidency University

Document Details

Submission ID

trn:oid:::1:3530740322

Submission Date

Apr 8, 2026, 3:08 PM GMT+5:30

Download Date

Apr 8, 2026, 3:16 PM GMT+5:30

File Name

CSG_16_Report.docx

File Size

7.6 MB

96 Pages

20,342 Words

120,861 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

AI-Driven Public Health Chatbot for Disease Awareness

A PROJECT REPORT

Submitted by

S. MOHAMMED AFFAN - 20221CSG0020

KARTHIK MP - 20221CSG0059

CHAITHRA C - 20221CSG0092

Under the guidance of,

Mr. Asif Ahmad Najar

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

PRESIDENCY UNIVERSITY

BENGALURU

APRIL 2026

PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "AI-DRIVEN PUBLIC HEALTH CHATBOT FOR DISEASE AWARNESS" is a Bonafide work of "S. MOHAMMED AFFAN (20221CSG0020), KARTHIK MP (20221CSG0059), CHAITHRA C (20221CSG0092)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND TECHNOLOGY, ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING during 2025-26.

Mr. Asif Ahmad Najar
Project Guide
PSCS
Presidency University

Ms Benitha Christinal J
Program Project Coordinator
PSCS
Presidency University

Dr. Sampath A K
Dr. Geetha A
School Project Coordinators
PSCS
Presidency University

Dr. Pallavi
Head of the Department
PSCS
Presidency University

Dr. Shakkeera L
Associate Dean
PSCS
Presidency University

Dr. Duraipandian N
Dean
PSCS & PSIS
Presidency University

Examiners

Sl. No.	Name	Signature	Date
1			

2			

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND TECHNOLOGY, ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING at Presidency University, Bengaluru, named S. MOHAMMED AFFAN, KARTHIK MP, CHAITHRA C , hereby declare that the project work titled **"AI-DRIVEN PUBLIC HEALTH CHATBOT FOR DISEASE AWARNESS"** has been independently carried out by us and submitted in partial fulfilment for the award of the degree of B.Tech in COMPUTER SCIENCE AND TECHNOLOGY (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

S. MOHAMMED AFFAN USN: 20221CSG0020

KARTHIK MP USN: 20221CSG0059

CHAITHRA C USN: 20221CSG0092

PLACE: BENGALURU

DATE: -April 2026

ACKNOWLEDGEMENT

For completing this project work, We/I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. We extend our gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the project.

I would like to sincerely thank my internal guide **Mr. Asif Ahmad Najar, Assistant Professor**, Presidency School of Computer Science and Engineering, Presidency University, for his moral support, motivation, timely guidance, and encouragement provided to us during the period of our project work.

I am also thankful to **Dr. Pallavi, Professor, Head of the Department, Presidency School of Computer Science and Engineering** Presidency University, for his mentorship and encouragement.

We express our cordial thanks to **Dr. Duraipandian N**, Dean PSCS & PSIS, **Dr. Shakkeera L**, Associate Dean, Presidency School of computer Science and Engineering and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my project work.

We are grateful to **Dr. Sampath A K, and Dr. Geetha A, PSCS** Project Coordinators**, Ms Benitha Christinal J, Program Project Coordinator**, Presidency School of Computer Science and Engineering, or facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

We are also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

S. MOHAMMED AFFAN

KARTHIK MP

CHAITHRA C

Abstract

Access to credible and up-to-date healthcare information is a huge hurdle for rural and semi-urban populations with scarce medical infrastructure, misinformation, and delayed communication of the outbreak which contributes to avoidable health risks. Existing digital health platforms are often dependent on high bandwidth, have limited conversational accessibility or deliver unverified responses as a result of uncontrolled language model outputs. In safety critical areas such as public health, having a structured, fact-based, and relevant medical information to regions to reduce misinformation and promote responsible behaviours in healthcare is vital.

This project is an Artificial-Intelligence based Public Health Chatbot that is designed to give awareness about preventive healthcare, education about diseases, information about drugs, the understanding of symptoms, and discovery of nearby hospitals via WhatsApp. The system is designed with a deterministic medical routing architecture that is combined with Retrieval-Augmented Generation (RAG). Medical knowledge is extracted from healthcare documents that have been curated, and converted into semantic embeddings which are stored in a FAISS vector database for similarity-based retrieval. A modular tool-based design makes user queries available to the specialized components - for example, a Symptom Analysis Engine, Drug Intelligence Module, Hospital Finder Tool (from Google places API) and a document-grounded medical knowledge engine. A safety and emergency detection layer provides non-diagnostic response and gives priority to the most urgent cases such as chest pain or difficulty in breathing. The chatbot is deployed by using fastapi backend services and twilio whatsapp integration which allow for accessible and real-time communication.

The system implemented is able to show successful structured medical query routing, context-aware response generation, emergency prioritization, and real-time hospital discovery in a single unified platform. By using both deterministic routing combined with RAG-based grounding, the assistant has less chances of hallucinated outputs and is more reliable when compared to standalone large language model approaches. The solution offers a framework that is accessible and scalable and constrained by safety and is able to improve public health awareness and facilitate responsible healthcare decision-making in underserved communities.

Keywords — Artificial Intelligence, Public Health Chatbot, Retrieval-Augmented Generation (RAG), Medical Question Answering, Symptom Analysis, Drug Information System, FAISS Vector Database, LangChain, FastAPI, WhatsApp Integration, Emergency Detection, Google Places API.

Table of Content

Sl. No.	Title	Page No.
	Declaration	
	Acknowledgement	
	Abstract	
	List of Figures	
	List of Tables	
	Abbreviations	
1.	Introduction 1.1 Background 1.2 Statistics of project 1.3 Prior existing technologies 1.4 Proposed approach 1.5 Objectives 1.6 SDGs 1.7 Overview of project report	
2.	Literature review 2.1 Literature Survey 2.2 Literature Gap Identification 2.3 Summary of Literatures Reviewed	
3.	Methodology 3.1 Agile Scrum Model 3.2 Overall System Development Workflow 3.3 Knowledge Base Preparation and Processing 3.4 RAG - Framework 3.5 Deterministic Routing Strategy 3.6 Safety and Emergency Detection Mechanism 3.7 Integration and Deployment Approach 3.8 Testing and Verification Strategy	
4.	Project management 4.1 Project timeline	

	4.1.1 Visualization of timeline 4.2 Risk analysis 4.3 Project budget 4.4 Resource Management 4.4.1 Resource Allocation 4.4.2 Project Monitoring and Tracking	
5.	Analysis and Design 5.1 Requirements 5.5.1 System Software Requirement 5.5.2 Data requirement 5.2 Block Diagram 5.3 System Flow Chart 5.4 Standards 5.5 Mapping with IoTWF reference model layers 5.6 Domain model specification 5.7 Communication model 5.8 IoT deployment level 5.9 Functional view 5.10 Mapping IoT deployment level with functional view 5.11 Operational view 5.12 Other Design	
6.	Hardware, Software and Simulation 6.1 Hardware 6.2 Software development tools 6.3 Software code 6.4 Simulation	
7.	Evaluation and Results 7.1 Test points 7.2 Test plan 7.3 Test result 7.4 Insights	
8.	Social, Legal, Ethical, Sustainability and Safety Aspects 8.1 Social aspects	

	8.2 Legal aspects 8.3 Ethical aspects 8.4 Sustainability aspects 8.5 Safety aspects	
9.	Conclusion 9.1 Future Work	
	References	
	Base Paper	
	Appendix Appendix A: Data Sheets Appendix B: Dataset Information Appendix C: GitHub Appendix D: Project Images Appendix E: Plagiarism Report Appendix F: Publications	

List of Figures

Figure	Caption	Page No.
Fig 1.1	System Architecture Diagram	
Fig 1.2	Application Programming Interface	
Fig 3.1	Agile Scrum Methodology	
Fig 3.2	System Workflow	
Fig 4.1	Gantt Chart	
Fig 5.1	Functional Block Diagram	
Fig 5.2	System Flow Chart	
Fig 5.3	Mapping with IoT World Forum Model	
Fig 5.4	Request-response Communication Model	
Fig 5.5	IoT Deployment	
Fig 7.1	Performance Metrics Graph	
Fig 7.2	Response Time Comparison	
Fig A.1	GitHub Repository	
Fig A.2	Chatbot type of desirable greetings and assisted queries over WhatsApp.	
Fig A.3	Example of symptom analysis response generated by the chatbot.	
Fig A.4	Nearby hospital discovery using location sharing through WhatsApp	

List of Tables

Figure	Caption	Page No.
Table 1.1	Proposed System Modules and Function	
Table 2.1	Summary of Literature Review	
Table 4.1	Project Planning timeline	
Table 4.2	Project Implementation Timeline	
Table 4.3	Pestle Analysis	
Table 4.4	Project Risk Matrix	
Table 4.5	Project Budget Estimation	
Table 5.1	Summarizing Requirements	
Table 5.4	Mapping Project Layers with IoTWF Reference Models	
Table 5.5	Description of Domain Model	
Table 5.6	Mapping Deployment Levels with Functional View	
Table 7.1	Test Points and Functional Units	
Table 7.2	Example User Queries and Chatbot response	
Table 7.3	Performance Evaluation Metrics	
Table 7.4	Comparison With Traditional Search Systems	

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
B.Tech	Bachelor of Technology
COVID-19	Coronavirus Disease 2019
DB	Database
FAISS	Facebook AI Similarity Search
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IoT	Internet of Things
IoTWF	Internet of Things World Forum
JSON	JavaScript Object Notation
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PSCS	Presidency School of Computer Science
QA	Question Answering
RAG	Retrieval-Augmented Generation
SDG	Sustainable Development Goal
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

Chapter 1
Introduction

Access to good and timely healthcare information is still a major hurdle especially in rural and semi-urban areas where medical infrastructure, professional consultation and preventive awareness are often limited. People in such areas often rely on informal advice or unverified sources on the internet, which leads to a higher potential for misinformation, late intervention and self-medication. With the ever-expanding penetration of mobile phone usage and the rapid adoption of messaging platform services like WhatsApp, conversational AI technologies can provide an opportunity to address this information gap by providing structured health guidance directly to users via familiar and accessible digital channels.

The idea of this project is to design and implement an AI-driven medical assistant that can give the understanding of symptoms, information about medicines, awareness about some diseases, and find the medical hospitals nearby by integrating WhatsApp. The system is developed as a safety aware document-aware public health chatbot with Retrieval-Augmented Generation (RAG) coupled with deterministic routing to specialized medical tools. By combining semantic document retrieval (using a FAISS vector database) and the generation of a controlled language model, the chatbot is able to ensure that the responses are context-aware, verified by medical information, and do not involve direct diagnosis or treatment prescription.

The demand for such a system is due to the growing healthcare awareness gaps and the exponential rate medical misinformation is spreading on digital platforms. Studies suggest that effective management of a sizable percentage of preventable diseases can be ensured through early identification of symptoms, compliance in getting vaccinated and timely consultation; however, a paucity of accessible and understandable information continues to be the impediment to preventative action. Additionally, generative AI models alone, even though they are very good at natural language processing, are prone to hallucinations and cannot necessarily provide factually correct medical advice if they are not anchored to reliable and correct sources.

Prior existing technologies such as rule-based healthcare chatbots and keyword-driven retrieval systems, due to their lack of contextual understanding and scalability, and pure LLM-based ones due to the safety concerns of unverified outputs. The proposed approach goes beyond these technologies by combining modular tool-based routing, vector-based semantic retrieval,

emergency detection mechanisms and real-time hospital discovery all in a unified and safety constrained medical assistant framework.

1.1 Background

The integration of conversational agents into healthcare systems has received a great deal of attention in the past several years because of their potential for better accessibility, patient engagement, and dissemination of health information. Healthcare chatbots are currently used for patient education, scheduling appointments, tracking symptoms and disease awareness campaigns. A systematic review from Laranjo et al. [1] discussed the possibility of conversational agents to improve healthcare accessibility and patient self-management, but many of the implementations can be limited in contextual understanding and reliability. As digital technologies continue to evolve, the interest in the use of Artificial Intelligence (AI) to deliver scalable public health communication solutions is increasing.

Public health awareness is especially important in rural and sem-urban population where limited healthcare infrastructure and low physician-patient ratio pose barriers to access to timely medical consultation. Studies have shown that preventive awareness, early symptom recognition, and adherence to vaccination has a significant effect on the reduction in disease burden and healthcare cost. Wilson and Marasoiu [3] emphasise that chatbots can be used as effective tools in public health outreach and disease education, particularly during large scale health campaigns. Similarly, Miner et al. [9] noted that the conversational AI systems were deployed extensively during the time of the Covid-19 pandemic, in order to disseminate verified information and counter misinformation. Regardless of these deployments, lots of the systems were based on pre-established rules or fixed information structures limiting flexibility and situational logic.

Recent developments in Large Language Models (LLMs) have been contributing to the ability to understand natural language and answer medical questions. Singhal et al. [5] showed that LLMs represent a significant amount of clinical knowledge and are competitive on medical benchmarks. However, generative models are subject to hallucination - factually incorrect but plausible responses - which can be hazardous in a healthcare application. Ji et al [6] identified hallucination as a critical concern in high-stakes areas such as medicine where accuracy and reliability are key. These challenges highlight the need for grounding the response of AI on verified medical documentation for trustworthiness.

Retrieval-Augmented Generation (RAG) has become a promising alternative that helps to mitigate hallucinations by the combination of document retrieval and generative language models. Lewis et al. [7] proposed the RAG framework, and it showed an improvement in factual consistency for knowledge-intensive tasks. Further research, including that of Izacard and Grave [8], indicated that approaches based on retrieval are superior to generative-only systems in question answering applications. However, most of the existing implementations are generic, and do not include structured medical routing, emergency detection, or real-time integration of healthcare services. The background of this project builds upon these advancements by integrating RAG, modular medical tools, and safety-aware response mechanisms into a unified AI-driven public health assistant tailored for accessible messaging platforms.

1.2 Statistics

Disparity in access to healthcare is still a major issue in India, especially in rural areas where a major percentage of the population is located. The population of India is also said to be almost 65% rural as indicated by the World Bank [11], and there is a relative lack of healthcare facilities and professionals in rural settlements. The Ministry of Health and Family Welfare Rural Health Statistics report indicates the lack of primary healthcare centers, community health centers, and medical professionals in a number of states [12]. Low ratios of doctors to populations and geographical constraints tend to postpone the early diagnosis and preventive measures thus putting the risk of complications in conditions, which in actual sense can be controlled, on the higher. Such loopholes highlight the importance of accessible informational systems in health on the basis of awareness.

Preventive health care is still of high importance in alleviating disease burden particularly in cases of communicable diseases, maternal health, and lifestyle disorders. World Health Organization estimates the number of people who can be prevented by early symptoms, vaccine compliance, and early seeking consultation to be a significant percentage of the disease burden in the world [13]. Immunization and disease control programs in India focus on creating awareness and disseminating information in time as the most important parts of the national public health strategy. Nonetheless, there is still a lack of awareness as the number of people reached is low and the dissemination of misinformation in non-formal digital networks. Meanwhile, the Indian population has become very fast in embracing digital. According to the

Telecom Regulatory Authority of India, the number of internet subscribers is more than 800 million and the penetration is growing even in the rural regions [14]. WhatsApp is among the most popular messaging services in the country, and this will be an effective health communication channel. Using messaging-based AI systems may thus be used to make sure that more people are accessed than traditional web-based portals. The system, through its capacity to combine the knowledge of public health and a conversational AI on the platform people are already used to, will address the issue of the healthcare information as well as trends in digital accessibility.

Moreover, false information regarding illnesses, drugs and countermeasures against epidemics have become a landmark issue in the recent past in response to public health. Research undertaken in the context of the COVID-19 pandemic revealed that fake news on online platforms virally spread and frequently resulted in unsafe self-medication and postponed visits to hospitals [9]. These tendencies point to the necessity of organized, confirmed, and paper-based AI systems that can provide valuable information and offer the maximum of safety. This local and digital health environment is directly met by the creation of a deterministic safety-constrained, WhatsApp-driven medical assistant.

1.3 Prior existing technologies

Development of chatbots and AI-assisted medical systems has been changing over several technological phases, starting with rule-based conversational systems. Initial healthcare chatbots were based on preset decision trees and matching keywords to generate output. These systems delivered predictable and controlled outputs but did not have any sense of context and flexibility. According to Laranjo et al. [1], in the past numerous early healthcare conversational agents were rule-based and performed only a few limited functions like appointment scheduling or a simple patient education. As much as these systems were safe because of deterministic logic, they could not manage different and complicated medical queries. With the development of the digital health systems, the information retrieval strategies that are based on key words were introduced. These systems enabled the users to search healthcare databases and access the relevant documents or informational data. Even though the retrieval-based systems enhanced the accessibility of the information as opposed to the non-interactive web portals, they lacked conversational interaction and contextual summarization. According to Bickmore et al. [2], conversational agents enhanced accessibility within a healthcare environment, but the safety of

responses and the factual validation were still a major issue. Equally, Wilson and Marasoiu [3] reported that chatbots in public health were often scripted and used shallow natural language processing systems, which reduced scalability and flexibility in real-world applications. As deep learning and large language models (LLM) advanced, medical chatbots that are generated by AI appeared.

Other researchers like Singhal et al. [5] were able to show that LLMs encode a significant amount of medical knowledge and achieve competitiveness in medical question-answering tasks. Such systems have high levels of natural language processing and are capable of producing fluent context-sensitive responses. Generative models are however more susceptible to hallucination, or production of plausible but false information. According to Ji et al. [6], a hallucination is one of the most severe issues of natural language generation systems, particularly those that are safety-related, such as healthcare. The generative aspect of standalone LLM systems is not under control and thus poses risks in terms of reliability when used directly in medical advisory tasks.

Retrieval-Augmented Generation (RAG) frameworks were proposed to deal with hallucination and the factuality of the generated text. Lewis et al. [7] offered RAG, which is a combination of document retrieval and generative models to improve the accuracy of knowledge-intensive activities. RAG limits the model to grounded context by retrieving the relevant documents prior to the generation of responses. Izacard and Grave [8] also showed that retrieval based augmentation is more effective in enhancing the performance on question answering than generative based methods. The current state of most RAG implementations is general-purpose, even though they are not yet structured medical routing, emergency prioritization, and coupled with real-time healthcare services.

AI-driven chatbots were actively used during the spread of health information and misinformation during the COVID-19 pandemic due to the common use of AI-powered chatbots during public health emergencies. Miner et al. [9] also reported the massive implementation of conversational AI tools in communicating and educating about outbreaks. Branda et al. [4] emphasized the importance of AI-based chatbots in the case of a health emergency and pointed to their benefit in disease awareness and prevention communication. Most of these systems were however rule-based, low context in their reasoning or had no formal grounding mechanisms to guarantee accuracy to responses. The necessity to address this gap in the previous technologies contributes to the creation of a modular, tool-based and retrieval-

based medical assistant that incorporates safety solutions and real-time healthcare assistance into an integrated structure.

1.4 Proposed approach

The proposed project is set to create a secure, organized, and accessible AI-based medical assistant, which can provide preventive healthcare awareness, symptom knowledge, drug information, and hospital discovery services through WhatsApp connectivity. The fundamental task is to deliver document-based, context-dependent, and safety-restricted replies and reduce the chances of misinformation that are usually prevalent in isolated generative artificial intelligence systems. It is also expected that the system will act as an educational and decision-support tool that will promote responsible healthcare behavior without substituting professional medical diagnosis.

This strategy is motivated by the confluence of the problem of gaps in healthcare accessibility and the use of previous chatbot technologies. Although rule based systems are not flexible and large language models have a risk of hallucinations, it is evident that there is a need to have a balanced system that provides intelligence and reliability. Also, since the broader population of people currently uses the messaging systems like WhatsApp, there is a possibility of conveying healthcare information straight into a commonly used communication system. The reason is to design a modular AI system that would be scaled and safe, especially in rural and semi-urban communities that have a high demand of convenient and dependable health awareness systems.

The suggested solution is based on a deterministic architecture of routing and Retrieval-Augmented Generation (RAG) and tool-based modules. User queries sent in by WhatsApp are converted into a FastAPI backend, and directed by a medical router module to a specialist tool, one of four tools (as shown by the arrows in Figure 1.1 (System Architecture Diagram)) to be used in Symptom Analysis Tool, Drug Intelligence Tool, Hospital Finder Tool, or Medical Knowledge RAG Engine. RAG component is based on document embedding, similarity search using the FAISS, and controlled language model generation that make sure the responses are based on the verified medical documents. A layer of emergency detection is a second layer, which identifies such life-threatening conditions as chest pain or difficulty breathing and directs users to direct help at the hospital.

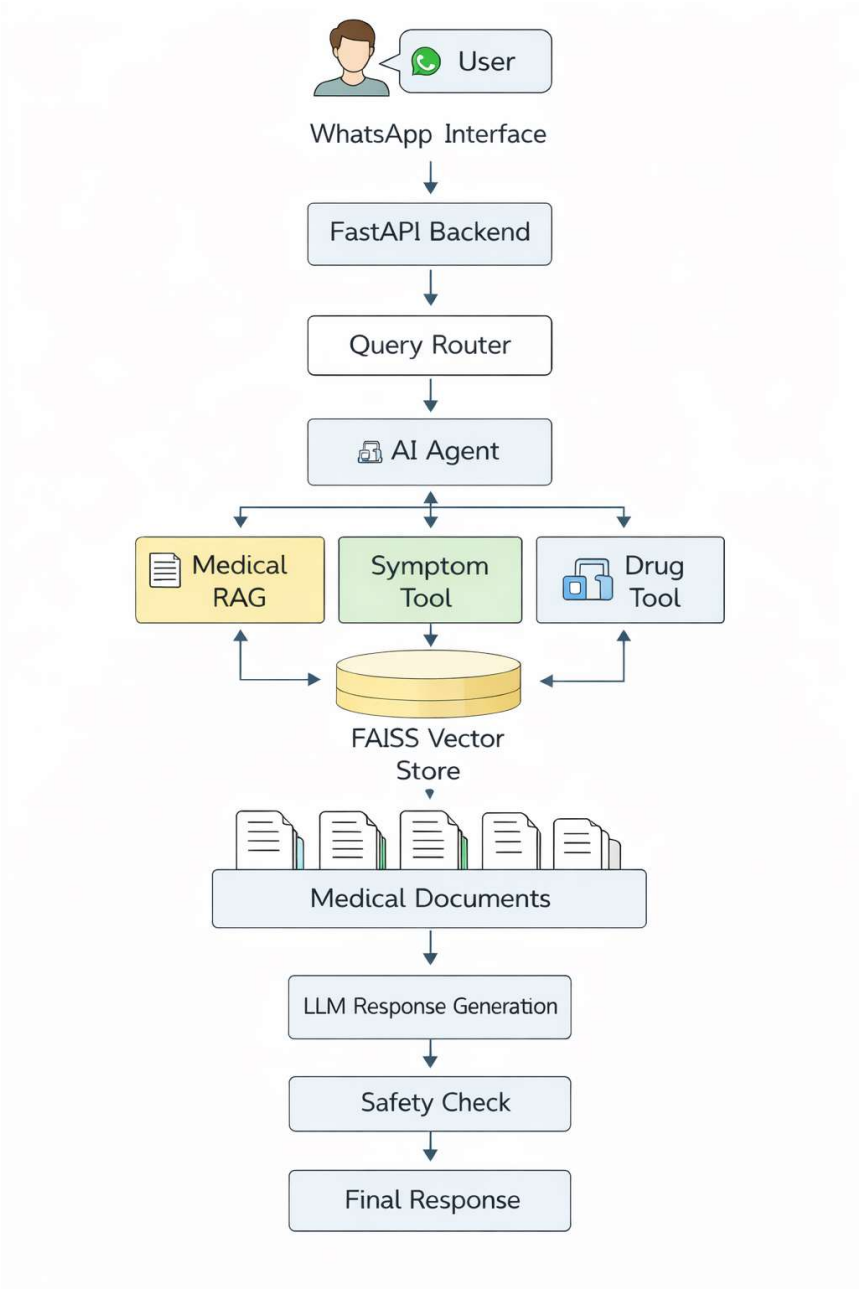


Fig 1.1 System Architecture Diagram

Table 1.1 (Proposed System Modules and Functions) gives a summary of system modules and system functions.

Table 1.1 Proposed System Modules and Functions

Module	Function	Description
Medical Router	Query Classification	Analyzes user input and routes the query to the appropriate module based on intent (symptom, drug, hospital, or general query).
Symptom Analysis Tool	Symptom Interpretation	Processes user-described symptoms and provides possible causes, basic remedies, and advice on when to seek medical help.
Drug Intelligence Tool	Drug Information Retrieval	Provides structured information about medicines including usage, side effects, and safety precautions.
Hospital Finder Tool	Location-Based Search	Uses Google Places API to identify nearby hospitals based on user location and provides relevant details.
Medical Knowledge RAG Engine	Contextual Response Generation	Retrieves relevant medical documents using FAISS and generates accurate, context-aware responses using a language model.
FAISS Vector Database	Semantic Search	Stores document embeddings and enables fast similarity-based retrieval for RAG processing.
Emergency Detection Layer	Risk Identification	Detects critical health conditions (e.g., chest pain, breathing issues) and prioritizes emergency guidance.
WhatsApp Interface (Twilio)	User Interaction	Enables real-time communication between users and the chatbot through WhatsApp messaging.
FastAPI Backend	Request Processing	Handles incoming requests, manages routing, integrates tools, and generates responses.

The uses of the proposed system are preventive healthcare education, the awareness of common diseases, safe medication instruction, emergency prioritization, the dissemination of information about vaccinations, and real-time aid with the location of the hospital. During the outbreak of a disease, the chatbot can be used as an additional public health awareness system and complement to government health communications. Although these are the benefits, there are some shortcomings. The system is based on the quality and coverage of the medical knowledge base, and it might need to keep on updating its documents in order to include new medical guidelines and it uses the internet to provide real-time location services. Moreover, the safety layer minimizes dangerous outputs, although the assistant cannot substitute professional diagnosis, which should always be used within specified non-clinical limits.

1.5 Objectives

Objective 1: Design and develop a WhatsApp-Based AI Medical Chatbot with Structured Query Routing

Conceive and develop an AI-based medical chatbot that uses WhatsApp that analyzes natural language user interaction and redirects these to predetermined healthcare domains like symptom identification, drugs information, hospital research, and medical research. The routing system must be deterministic on the intent of the user input and invoke suitable backend tools instead of depending on just free-form generative art. The system has to show proper and correct categorization of queries at runtime testing.

Objective 2: To construct a Retrieval-Augmented Generation (RAG) Based Medical Knowledge Engine.

Deploy a RAG (with document embedding methods and FAISS) approach of semantically relevant medical content to curated healthcare documents. Fine-tune the content retrieved to the controlled Large Language Model and produce context-grounded responses. The system is supposed to make sure that responses are only produced using retrieved evidence and that they reduce hallucinated outputs during the evaluation testing.

Objective 3: Implement an Emergency Detection and Safety Validation Module

Build a safety layer that will be able to recognize dangerous medical questions like the chest pain, symptoms of a stroke, and difficulty in breathing. This module must take control of the normal response generation and give priority to urgent medical advice and hospital discovery.

Also, put in place response filtering systems to avert diagnostic claims, prescription recommendations or unhealthy medical advice. When query processing is done in real-time, the safety module should be activated.

Objective 4: Develop a Modular and Scalable Backend System Architecture

An architecture Design and build a modular backend around FastAPI and tool-based orchestration logic to coordinate autonomous units like symptom analysis, drug intelligence, hospital finder, and RAG engine. Maintainability, scalability, and separation of concerns of the system should be achieved by organising the code. The architecture should be in a way that it can easily support the integration of new medical tools without redesigning the whole system.

Objective 5: Implement AI Medical Assistant through WhatsApp.

Integrate Twilio WhatsApp API and Google Places API to allow real-time interaction with the user and locate a hospital based on the location. The system must be able to process user-messages, webhook-messages and provide structured hospital information such as name, address and rating. The deployment should prove end to end operational capacity in live messaging conditions.

1.6 SDGs

The United Nations Sustainable Development Goals (SDGs), as shown in Figure 1.1, offers an international strategy in order to solve urgent social, economic, and environmental issues in the 2030 Agenda of Sustainable Development [15]. The suggested AI-based Public Health Medical Assistant is in line with several SDGs, especially with health, equity, and digital innovation.

SDG 3: Good Health and Well-Being

The main orientation of this project is that of SDG 3: Good Health and Well-Being, which strives to provide healthy living and wellbeing to all people of all ages [16]. This objective is facilitated by the system through preventive healthcare awareness, early symptoms advice, medication education, and emergency prioritization systems.

The project is donating to enhance universal health coverage awareness and early medical intervention, particularly in underserved community by facilitating easy access to health education via WhatsApp.

SDG 10: Reduced Inequalities

The SDG 10: Reduced Inequalities that also focuses on decreasing inequalities between and among people also correlates with the project [17]. Rural and semi-urban populations usually have inadequate access to medical providers and reputable medical knowledge. The chatbot will be used to close informational inequality and enhance digital health among a wide range of socio-economic groups by enhancing health information through the use of widely used messaging platforms, which are structured and document-based.

SDG 9: Industry, Innovation and Infrastructure

Moreover, the combination of Artificial Intelligence, semantic access systems, and digital health infrastructure promotes SDG 9: Industry, Innovation and Infrastructure [18]. The project illustrates application of new AI-based technologies to create robust digital health communication infrastructure, including Retrieval-Augmented Generation, vectors database and conversational agents. The system allows the development of sustainable and inclusive technological innovation by integrating AI tools of modern technologies and deploying messages at scale.

Through alignment with these SDGs, the proposed medical assistant not only serves as a technical solution but also supports broader global objectives aimed at improving health accessibility, reducing inequalities, and promoting responsible technological innovation.

Fig 1.2 Sustainable development goals [15]

1.7 Overview of project report

In this report, the conceptualization and testing of the AI-Driven Public Health Medical Assistant, a WhatsApp-enabled, Retrieval-Augmented Generation (RAG) based healthcare chatbot, is described that delivers safe and organized, document-based medical advice.

Chapter 1 provides a preliminary discussion of the project, namely overviewing the problem of healthcare in care, statistical rationale, existing technologies, the project approach, formulated purpose, and alignment to Sustainable Development Goals (SDGs). This chapter predetermines the motivation and preconditions of the system.

The Literature Review is introduced in Chapter 2 and examines the previous studies on healthcare chatbots, conversational AI systems, risks of hallucinating in large language models, and retrieval-augmented methods. This chapter presents the current limitations and outlines the gap in the research that the proposed system will be able to fill.

Chapter 3 thereafter describes the Methodology used, the system workflow, which consists of knowledge-base preparation, embedding generation, storing vectors, deterministic routing, safety implementation and deployment using WhatsApp.

Chapter 4 targets the Project Management elements that include planning, structuring of timeline, risk identification and budgeting concerns.

Chapter 5 is the system Analysis and Design, which explains the architectural structure, elements of the system, and data flow.

Chapter 6 is about the environment of the implementation, hardware, software tools, and significant code modules.

Chapter 7 will assess the performance of the systems by conducting formal testing and analysis of the results.

Chapter 8 will assess the social, legal, ethical, sustainability and safety ramifications of implementing AI in healthcare. The report will be summarized by the summary of successes, the limitations, and the future improvements.

Chapter 2

Literature review

The high pace of conversational artificial intelligence and large language model growth has had a profound effect on the creation of healthcare support systems. Over the last couple of years, scholars have examined the application of chatbots, intelligent virtual assistants, and retrieval-based systems to provide better patient engagement, automate health communication, and increase access to medical information. Nevertheless, healthcare is a safety-sensitive field where reliability, fact base and ethical concerns are supreme. Although the first healthcare chatbots were more rule-based and had low levels of contextual knowledge, new methods use deep-learning-based methods and large language models that generate human-like responses. Regardless of these advances, problems like hallucination, absence of contextual presentation and risks to safety remain, especially in clinical or community health development. To solve these problems, the Retrieval-Augmented Generation (RAG) and document-grounded systems have been proposed to enhance the consistency of facts and minimize misinformation. The reviewed literature in this chapter discusses some of the major contributions in healthcare conversational agents, AI-driven medical question answering, hallucinations in generative models, and retrieval-enhanced generation methods. This review defines the gap in research, which leads to the creation of a modular, safety-conscious, WhatsApp-enabled public health chatbot that combines deterministic routing and document-grounded response generation, based on the analysis of the concepts, methodologies, results, and limitations of these works.

2.1 Literature Survey

[1] Laranjo, L., Dunn, A. G., Tong, H. L., Kocaballi, A. B., Chen, J., Bashir, R., Surian, D., Gallego, B., Magrabi, F., Lau, A. Y. S., & Coiera, E., "Conversational agents in healthcare: A systematic review," *Journal of the American Medical Informatics Association*, vol. 25, no. 9, pp. 1248–1258, 2018.

Laranjo et al. provided a systematic review of the application of conversational agents in healthcare environments. The review examined the literature on the application of chatbots to educating patients, supporting them mentally, managing their lifestyles, and monitoring chronic diseases. The authors divided chatbot applications according to functionality, automation degree, and interaction design and characterized rule-based systems and more

complex AI-based conversational models. The review noted that the available healthcare chatbots at the time were mostly based on predetermined scripts and decision-tree reasoning and not on modern machine learning methods of natural language processing. Although such systems have enhanced accessibility and interaction with patients, particularly with regard to giving health education and behavioral support, most of them did not have rigorous validation and standardized measures of evaluation. Small sample sizes, brief timeframes of evaluation, and little safety considerations were identified as limitations of the study. Moreover, not many systems were shown to be capable of scaled deployment to large populations or connect to real-time sources of health data. The authors noted that better methodological frameworks, metrics of evaluation, and more sophisticated natural language process approaches are required to promote the reliability and trustworthiness of chatbots in a healthcare setting. The results of the given review represent the background knowledge of the current state of chatbots in the early healthcare setting and the gap in scientific research concerning the creation of the context-aware, scalable, and safety-constrained AI-guided assistant in healthcare. The identified gap is the main reason behind the implementation of skeleton routing systems and document-based generation of the proposed public health chatbot system [1].

[2] Bickmore, T. W., Trinh, H., Asadi, R., & Olafsson, S., "Safety First: Conversational Agents for Health Information," *Journal of Biomedical Informatics*, vol. 82, pp. 1–9, 2018.

Bickmore et al. researched the design and safety concepts of conversational agents which were designed to provide health information to users. Special attention was given to the study because the conversational systems should not deliver harm-giving or misleading medical advice. The authors evaluated risks of automated health communication particularly where the users consider chatbot advice as a clinical diagnosis. Their activities focused on the control of dialogue structure, pre-established safety limitations, and on the necessity of medical scope restriction in conversational systems. The researchers emphasised that conversational agents can enhance health literacy and user engagement but only in cases where they are designed to be safe, otherwise such tools can be ethically and medically dangerous. One important contribution of the work was the description of the guardrail strategies, i.e. fallback responses, clear disclaimers and escalation procedures in the high-risk situations. In the paper, however, the main issues addressed were design frameworks, and not technical processes of integrating real-time knowledge validation, or document-grounded generation. Although the concept of

safety was highlighted, the research failed to offer a computational design to reduce hallucinations of AI-based systems. This shortcoming encourages the adoption of a deterministic routing layer and safety validation module in the intended public health chatbot in order to make findings medically responsible [2].

[3] Wilson, L., & Marasoiu, M., "The Development and Use of Chatbots in Public Health: Scoping Review," *JMIR Human Factors*, vol. 9, no. 4, e35882, 2022.

Wilson and Marasoiu performed a scoping review of the literature on the application of chatbots in public health interventions, comprising of disease awareness, vaccination campaigns, and outbreak communications. The paper has explored the different chatbot applications in different nations and evaluated their effectiveness in enhancing their accessibility and user interaction. The authors noted that chatbots used in public health can be helpful especially when the awareness work is conducted on a large scale and emergencies are being handled since most of them are scalable, and the information is delivered fast. Nonetheless, the majority of reviewed systems were based on scripting with rules and did not have sophisticated Natural Language Understanding. Some of the challenges that were identified in the paper included the lack of personalization, the lack of proper evaluation metrics, and the failure to address complex queries by the user. Also, it was noted that AI-driven systems can raise the issue of data privacy, transparency, and ethical concerns. However, even though the review admitted the possibilities of AI-enhanced conversational agents, it stated that not many systems used document-grounded or retrieval-based methods to achieve factual accuracy. This brings into the limelight the necessity to incorporate the best semantic retrieval and AI grounding systems, which are taken up in the proposed system by Retrieval-Augmented Generation and modular routing architecture [3].

[4] Branda, F., Stella, M., Ceccarelli, C., Cabitza, F., Ceccarelli, G., Maruotti, A., Ciccozzi, M., & Scarpa, F., "The Role of AI-Based Chatbots in Public Health Emergencies: A Narrative Review," *Future Internet*, vol. 17, no. 4, 145, 2025.

Branda et al. examined the issue of using AI-based chatbots in the context of emergent situations in the field of public health, specifically in the case of outbreaks of infectious diseases. The present study examined the implementation of chatbots in the context of crisis

times and how their support positively influenced dissemination of information in real time, management of rumors, and behavioral advice. The authors emphasized that speed of information delivery in emergency situations is of high importance and pointed out that AI systems can help to decrease misinformation and panic. Nevertheless, the review has also highlighted that there are major technological issues such as the reliability of the responses, real-time updates of the data and filtering of misinformation. A significant number of deployed systems were rule-based or not structured in terms of evidence-grounding, and hence susceptible to information inconsistency. Moreover, the authors emphasized the necessity of incorporating proven sources of data and being trustworthy in the process of crisis communication. Although the paper encompasses the importance of AI chatbots during the emergency of a public health crisis, it is not considering the technical framework of ensuring document-backed responses in detail. This constraint provides the topicality of the offered project, where the relevance of the FAISS-based retrieval and safety restrictions are introduced to guarantee the trustful information spread in the case of the potential outbreaks [4].

[5] Singhal, K., Tu, T., Gottweis, J., Sayres, R., Wulczyn, E., Hou, L., Clark, K., Pfohl, S., Cole-Lewis, H., Neal, D., et al., "Large Language Models Encode Clinical Knowledge," *Nature*, vol. 620, pp. 172–180, 2023.

Singhal et al. studied the ability of large language models (LLMs) to encode and provide clinical knowledge by benchmark-based evaluation. The research presented a domain-specific model in medicine and its performance in a number of medical question-answering tasks. Findings indicated that state of the art LLMs are capable of high accuracy when used on standardized medical examination data, which is attributed to good contextual reasoning and language comprehending. The authors noted the possibility of LLMs in the area of clinical documentation, medical education, and decision-support application. Nonetheless, the paper also recognized the weaknesses (such as the possibility of hallucinations, the training data that might be biased, and inability to explain generated outputs). The LLM responses were probabilistic, which is quite high in benchmark performance, and could generate false or unprovable medical statements. These results highlight that, without external grounding mechanisms, standalone generative systems cannot be used in safety-critical healthcare applications despite having large amounts of medical knowledge in the form of LLMs. It is a direct signal to the methodological decision in the proposed system to blend generative AI with

retrieval-based grounding, deterministic routing, and safety constraints to achieve better reliability and factual consistency in communicating in the field of public health [5].

[6] Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P., "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.

The analysis of hallucination in natural language generation systems was carried out in a large-scale survey by Ji et al. The article divides hallucinations into intrinsic and extrinsic and discusses the causes of such hallucinations with reference to various architectures, such as transformer-based large language models. The authors study the evaluation metrics to identify a hallucination and comment on the mitigation strategies, including reinforcement learning, retrieval grounding, and fact-checking mechanisms. The researchers highlight the fact that the risks of hallucinations are more pronounced in the high-stakes areas, including healthcare, law, and finance, where the precision of facts is paramount. Although some of the mitigation strategies were researched, the survey suggests that none of these techniques is effective in eliminating hallucinations in generative models. Rather, hybrid methods in which external knowledge is retrieved and controlled generation is performed seem to be most promising. One of the weaknesses that were observed in the article is that there are no universally recognized measurement criteria of hallucination in field-specific cases such as medicine. The information received during this survey is a strong reason to consider the implementation of Retrieval-Augmented Generation in the proposed project, where the external document grounding with the help of semantic search with FAISS is adopted to minimize the likelihood of generated or unsubstantiated medical answers [6].

[7] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

Retrieval-Augmented Generation in Knowledge-Intensive NLP Tasks. Piktus, A., Perez, E., Piktus, A., Perez, E., Petroni, F., Karpukhin, V., Goyal, N., Kuehler, H., Lewis, M., Yih, W., Rocktaschel, T., Riedel, S., and Kiela, D., Advances in Neural Information Processing Systems

(NeurIPS), vol. 33, 94599474, 2 A hybrid model, Retrieval-Augmented Generation (RAG), proposed by Lewis et al. is a combination of dense document retrieval and generative sequence-to-sequence models. The given strategy brings the pertinent documents in a large body of knowledge and trains the language model on the retrieved context when generating a response. The authors established that the RAG performs better than all-parametric models on knowledge-based tests like open-domain question answering. As the framework is based on retrieved material, it enhances accuracy in factual responses and yet the generative flexibility. Nevertheless, the original implementation of RAG was tested mainly on non-domain-specific datasets, but not on domain-specific ones, like in the healthcare context. Also, the quality and coverage of the retrieval corpus is hugely important to the system performance. Regardless of these shortcomings, the RAG framework offers a structural framework of integrating retrieval and generation well. The proposed public health chatbot directly builds upon this framework by implementing FAISS-based vector retrieval and contextual generation restricted to curated medical documents, thereby adapting RAG to a safety-critical healthcare domain [7].

[8] Izacard, G., & Grave, E., "Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering," ***Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*****, pp. 874–880, 2021.**

The article by Izacard and Grave discussed how passage retrieval can be combined with generative transformer models to enhance open-domain question answering. They employ dense passage retrieval to identify the relevant document passages and use them in direct encoder-decoder architecture of the generative model. Factors such as factual accuracy and relevance of the answer were improved experimentally in relation to standalone generative systems. The authors proved that retrieval and generation are better combined so that models base on less memorized knowledge and more external evidence. However, the analysis was on benchmark QA datasets and not on safety-sensitive applications in reality. The paper ignored field related limitations including medical ethics, emergency prioritization, or control of misinformation. Although it is technically sound, the solution does not have application-layer protections, with the conditions of a healthcare facility. The suggested project builds on this retrieval-based idea with the exception of deterministic routing, emergency detection, and safety filtering in association with semantic retrieval, thus, modifying retrieval-enhanced generation to accountable medical information delivery [8].

[9] Miner, A. S., Laranjo, L., & Kocaballi, A. B., "Chatbots in the Fight Against the COVID-19 Pandemic," *npj Digital Medicine*, vol. 3, no. 65, pp. 1–4, 2020.

Miner et al. overviewed the use of chatbots during the COVID-19 pandemic with references to their use to spread information about the current state of affairs on the topic of public health, screening of symptoms, and misinformation. The paper highlights the fact that conversational agents could be scaled very fast to offer real time information in the event of a worldwide crisis. Such systems were efficient to handle high traffic of queries as well as to decrease the load on healthcare helplines. Nevertheless, the authors have also mentioned that a significant number of pandemic chatbots were rule-based, and they often had only scripted interactions and could not react dynamically to complex or unanticipated user queries. Furthermore, quality of responses was different on the basis of system design and integration with tested data sources. Whereas chatbots were promising in the communication of public health, their reliability and contextual adaptability were an issue. It supports the necessity of the communication of the retrieval-grounded AI mechanisms and organized routing strategies, used in the suggested system to increase reliability and flexibility in the situations of public health [9].

[10] Devaraj, A., Sharma, A., & Ghosh, P., "Evidence-Based Clinical Decision Support Using Natural Language Processing and Retrieval Systems," *Journal of Biomedical Informatics*, vol. 128, 104002, 2022.

Devaraj et al. suggested a document-based clinical decision support system built upon the natural language processing and retrieval processes to deliver evidence-based medical knowledge. Their system combines the retrieval of medical literature with dedicated pipelines to process the information and produce the results based on authoritative sources as opposed to generative approximation. It was revealed that the study had a high level of reliability and trust in comparison to the conventional AI-based answer generating mechanisms. The authors highlighted that clinician confidence in AI-assisted decision aids can be boosted using evidence-based retrieval. Nevertheless, their use was rather specific to clinical specialists and did not have conversational interfaces that could be used by the general population. Moreover, the system did not include real-time messages deployment and emergency routing. Though it was good in document-grounded methodology, it was not accessibility-oriented. The suggested project builds on this idea and uses document-backed decision-support logic to turn it into a

conversational assistant on WhatsApp that is specifically built to support public awareness of health, including routing and safety restrictions done through tools [10].

2.2 Literature Gap Identification and Research Contribution

The literature review demonstrates that tremendous progress has been made in healthcare chatbots, conversational agents, and retrieval-augmented AI systems, though, there are still a number of research gaps. The first generation of chatbots in healthcare was mostly rule-based, restricting the contextual scalability and flexibility [1][3]. Despite the fact that these systems enhanced accessibility and engagement among users, they did not have high-level natural language recognition and powerful evaluation systems. More recent papers presented large language models that can give responses with fluent and context-awareness [5]; but, the risks of hallucination and factual inconsistency are significant issues in safety-critical medical settings [6]. Although retrieval-augmented generation models were shown to enhance the factual grounding [7][8], the vast majority of models were evaluated in general-domain question answering, and lacked healthcare-specific safety controls, emergency prioritization reasoning, and query routing. Moreover, chatbots used in the field of public health in case of emergency were either rule-based or had no deterministic control over the generation of responses, which restricted reliability and trust [4][9].

Moreover, the current document-based clinical systems were mainly designed to assist medical workers as opposed to making a public interface readily available [10]. There is scarcely a system that integrates routing based on modular tools, semantic tool recovering, real-time hospital discovery, and safety validation, into a single conversational system. It is still missing in bridging the gap between Retrieval-Augmented Generation and deterministic routing architectures specific to the case of public health awareness in rural and semi-urban settings. This study is aimed at filling this gap by building on a WhatsApp-enabled, safety conscious, modular AI medical assistant with document-grounded generation, emergency detection systems, structured tool coordination, and real-time service coordination. The purpose of this approach is to increase the reliability, scalability and accessibility and reduce the risk of misinformation during healthcare communication.

2.3 Summary of Literatures reviewed

As the reviewed literature in this chapter has shown, conversational AI has a great potential to enhance healthcare accessibility and communication with the population. Chatbots that used rules provided a better interaction experience, but were not as contextual; large language models were also more fluent, but there was a risk of hallucination. RA- Generation frameworks offered a better factual basis but did not have extensive modification to safety relevant healthcare purposes. The need towards scalability and speedy dissemination of information as a result of public health deployments commented on systems weaknesses in deterministic routing and safety enforcement. Together, these works determine the need to have a hybrid system that incorporates retrieval-based grounding, the modular tool orchestration, emergency detection, and safe deployment within a real-world messaging platform - the basis of the proposed AI-driven public health medical assistant.

Table 2.1 Summary of Literature reviews

Ref. No.	Authors & Year	Focus Area	Approach Used	Key Contribution	Identified Gap
[1]	Laranjo et al., 2018	Healthcare conversational agents	Systematic review of rule-based and AI chatbots	Validated role of chatbots in patient education and engagement	Limited contextual intelligence and weak evaluation standards
[2]	Bickmore et al., 2018	Safety in health chatbots	Design-focused conversational frameworks	Emphasized need for safety guardrails and non-diagnostic responses	No technical grounding mechanism for response validation
[3]	Wilson & Marasoiu, 2022	Public health chatbot deployments	Scoping review	Demonstrated scalability for health awareness campaigns	Mostly rule-based, limited NLP sophistication
[4]	Branda et al., 2025	AI chatbots in emergencies	Narrative review	Highlighted outbreak communication effectiveness	Lack of structured factual grounding and reliability control
[5]	Singhal et al., 2023	Medical knowledge in LLMs	Evaluation of medical LLM performance	Demonstrated strong clinical reasoning capability	Hallucination and unverifiable outputs remain concerns
[6]	Ji et al., 2023	Hallucination in NLP	Survey analysis	Identified hallucination types and mitigation techniques	No universal mitigation for healthcare-critical contexts
[7]	Lewis et al., 2020	Retrieval-Augmented Generation	Dense retrieval + generative model	Introduced RAG framework improving factual grounding	General-domain focus, not healthcare-specific
[8]	Izacard & Grave, 2021	Retrieval-enhanced QA	Passage retrieval + encoder-decoder	Improved answer accuracy using retrieval grounding	Lacked healthcare safety constraints
[9]	Miner et al., 2020	Pandemic chatbots	Review of COVID-19 chatbot deployments	Demonstrated scalability during crisis communication	Mostly scripted systems with limited adaptive reasoning
[10]	Devaraj et al., 2022	Evidence-based decision support	NLP + document retrieval system	Promoted document-backed AI support	No conversational public-facing interface

Chapter 3

Methodology

3.1 Methodology – Agile Scrum Model

The AI-Driven Public Health Medical Assistant is a proposed project, which was created through the Agile Scrum approach, an iterative and incremental model of software development, which is also popular in working with complex and changing projects [19]. Contrary to the linear models, e.g. Waterfall or V-Model, Agile allows to plan adaptively, refine and integrate feedback progressively. As the project implied several changing modules such as the Retrieval-Augmented Generation (RAG) engine, deterministic routing mechanism, safety validation layer, hospital finder tool, and WhatsApp integration, the iterative approach that includes the modular development strategy was more suitable than the strict sequential model.

The Agile structure as seen in Fig 3.1 Agile Scrum Methodology is made up of requirement analysis, sprint planning, system design, development, testing, review and iterative refinement. This project development lifecycle was planned in terms of structured sprints, and each of them was devoted to the implementation of certain functional modules. The first sprint was focused on the identification of the requirements and the definition of the backlog including the analysis of the public health issue, literature review, the formulation of system objectives, and specification of features including analysis of symptoms, drug information retrieval, emergency detection, and hospital discovery based on the location. This was captured in project backlog and ranked to be developed.

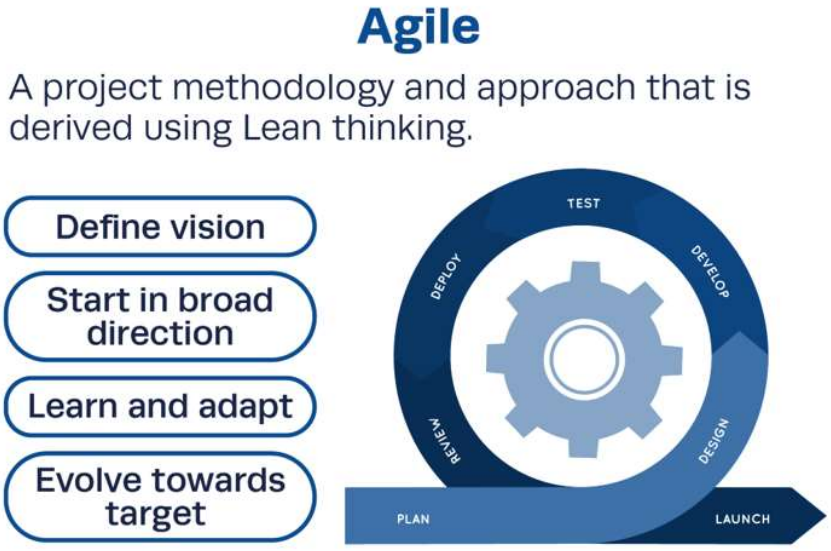


Fig 3.1 Agile Scrum Methodology [19]

The system design and functional implementation were handled in the later sprints. System architecture was designed on the high level, as well as WhatsApp integration pipeline, deterministic routing module, FAISS-based vector storage, Retrieval-Augmented Generation workflow, and safety implementation mechanisms. They formed functional modules which were independent units- the Symptom Tool, Drug Intelligence Tool, Hospital Finder Tool and Medical RAG Engine, to guarantee modular scaling and unit-level verification. The back-end services were implemented with FastAPI, LangChain gave orchestration, HuggingFace generated models, FAISS semantic vectors storage, and Twilio and Google Places APIs deployed and real-time services respectively.

The sprints were also followed by formal testing, unit testing, integration testing, verification and validation. Unit testing was done to make sure that individual modules worked, routing logic checks, and technology checks of the vector retrieval, emergency keyword detection, and check of API response. End-to-end query process, consistency of tool invocation, and context-grounded response generation was validated with the help of integration testing. Verification was done to ensure the system met predetermined technical requirements, including deterministic routing and safe enforcement of responses, and verification to real-world interaction scenarios to ensure stability, accuracy and compliance to safety. The reason behind Agile methodology selection is that it facilitates refinement that is iterative, development of modular architecture, quick improvement cycles, and ability to adapt to changes in AI system development, which were all aimed at creating a scalable and safety-conscious public health chatbot.

3.2 Overall System Development Workflow

The AI-Driven Public Health Medical Assistant was created in a guided technical process that has been initiated with the preparation of the knowledge base and completed with the launching of the application via a real-time messaging system. The modularity, reliability, safety enforcement, and smooth integration of conversational AI with organized medical information sources were ensured in the workflow. Fig 3.2 Overall System Development Workflow Fig 3.2 shows in a conceptual manner the successive interaction between data preparation, semantic retrieval, routing logic, safety enforcement and deployment layers.

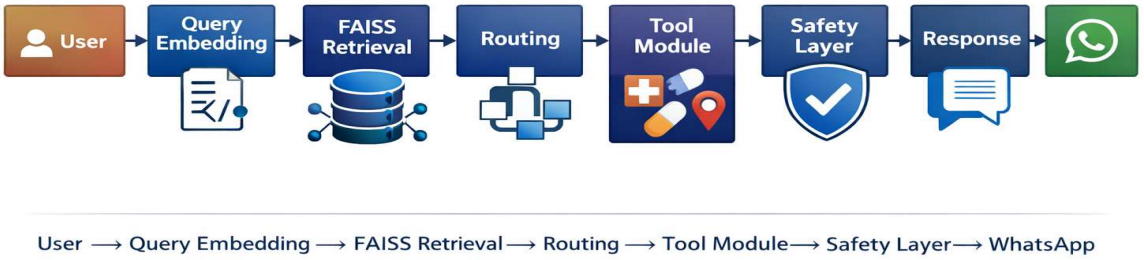


Fig 3.2 Overall System Development Workflow

The medical knowledge base preparation was the initial part of the workflow. Authoritative sources of medical documents were chosen to be used as the sources of authoritative content and were transformed using the text extraction and semantic segmentation methods. The text extracted was separated into context-sensitive bits to ensure maximum similarity search. Numerical embedding models were used to convert every text chunk into numerical vectors, in which case they have been indexed with FAISS (Facebook AI Similarity Search) to enable the rapid retrieval of semantics. This knowledge repository is the base of the chatbot system and is referred to as this vectorstore.

The second phase was that of applying the Retrieval-Augmented Generation (RAG) framework. In case a query is entered by a user via WhatsApp, it is received and translated into an embedding representation. FAISS uses semantic similarity search in order to extract the most relevant document fragments. Such retrieved segments are given as context input to the language model and allow response generation that is context-based. The use of the language model, which is restricted to the evidence retrieved minimizes the risks of hallucinations and guarantees factual consistency in the answers.

Deterministic routing and modular tool invocation are added to the third step of the workflow. The routing module analyses the queries made by the user and categorizes intent into classes like symptom analysis, drug information, hospital finder or general medical knowledge. Depending on the classified intent, the right tool module is called. This routing mechanism eliminates the unnecessary generative reactions and implies the systematic treatment of domain-specific tasks. Also, an emergency detection and safety validation system constantly analyzes the inquiries of the user to determine dangerous situations and successively ignore regular responses when immediate advice is necessary.

The last phase is the backend processing and deployment integration. FastAPI is the fundamental back-end platform that handles the requests and responses in HTTP, tool

management and the form of responses. Twilio WhatsApp API is to be used to help with the real-time communication with users, and the use of Google Places API can allow locating a hospital in question. When the messages are generated and filtered by the safety net, the completed message is sent back to the user through WhatsApp, and the process of interaction is closed. Such a structured workflow provides information delivery reliability, scalable architecture and safe deployment in real world.

3.3 Knowledge Base Preparation and Preprocessing

The quality and structure of the knowledge base in Retrieval- Augmented Generation (RAG) system is crucial to its effectiveness. In the suggested AI-Driven Public Health Medical Assistant the proposed selection of curated medical documents was the reliable source of authoritative information to provide reliability and factual response consistency. Such documents were filtered to eliminate irrelevant, ambiguous content and to make them close to preventive healthcare, symptoms of diseases, drug information, and overall preventive information about the health to the general population. Preprocessing entailed the creation of textual content of medical PDF files in the initial phase. The text was broken down using document parsing methods to convert structured and semi-structured medical information to machine-readable text. The importance was also made on maintaining semantic continuity at the expense of any formatting artifacts like headers, page numbers, and non-informative symbols. This move served to guarantee that extracted text had contextual meaning that could be taken to the semantics processing stage.

After extracting the text, it was divided into smaller semantic units of the documents. Chunking was aimed at keeping contextual coherence in each part rather than splitting it arbitrarily. An optimal chunk size was chosen to be able to retain enough context to make it meaningful to retrieve, but also allow maximizing the number of similarities searched. Excessively large chunks can make relevance be diluted and any smaller bits can discontinue flow of context. Hence, the parameters of segmentation were selected to ensure a good trade-off between computational efficiency and retrieval accuracy. The individual text fragments were then transformed into high-dimensional viewpoints under embedding models that were already trained. These embeddings learn semantic connections among pieces of texts that allow finding similar words without just matching the keywords. Embeddings generated were indexed with FAISS (Facebook AI Similarity Search) library to generate a permanent database of vectors

that facilitate a nearest-neighbour search. It is a kind of knowledge warehouse over which the chatbot is built to support context-based answers and minimize reliance on parametric memory of the language models.

The knowledge base preparation phase, through structured preprocessing, semantic chunking and vector indexing, makes sure that the system is able to extract the relevant medical information within an accurate and efficient method when engaged in a real time interaction.

3.4 Retrieval-Augmented Generation (RAG) Framework

The AI-Driven Public Health Medical Assistant follows a Retrieval-Augmented Generation (RAG) model to make sure the responses generated have an underpinning in the verified medical documentation and are not based on parametric knowledge of large language models. The conventional generative systems rely on internal learned representations and the results can be a hallucination or proved false in safety-related areas like healthcare. To overcome such limitation, the suggested system combines semantic retrieval and controlled language generation.

The RAG pipeline starts when a client places a query using WhatsApp. The incoming query is then transformed to a high-dimensional embedding representation by the same embedding model used in the preparation of the knowledge base. This is to provide semantic consistency in semantic space between document embeddings stored and query inputs. The query within this query is then passed through the FAISS vector index, which effectively searches in a similarity manner to be able to bring back to the knowledge base the top-k semantically-relevant document fragments.

The textual fragments that have been recalled are then given as contextual input to the language model. The language model generates answers instead of freely generating responses by being limited to formulating answers using the contextual evidence that was retrieved. The process improves facts consistency and drastically lowers the likelihood of hallucinating. The value of the parameter top-k was determined to have sufficient information on the model without introducing irrelevant information, and to have enough richness of context.

Also, response formatting policies were established in order to have structured medical advice. The output generated is limited to give informational advice like potential causes, general prevention measures and advice to consult professional medical attention where appropriate.

RAG framework does not permit direct prescription and diagnosis claims, which ensures responsible AI use within non-clinical parameters.

RAG provides the necessary intelligence layer upon which the chatbot operates, which is based on semantic retrieval, contextual grounding, and controlled production and allows delivery of information with reliability, scalability, and medical responsibility.

3.5 Deterministic Routing Strategy

Unlike the case of generic conversational agents, which do not have a routing mechanism and only use generative models to answer all types of queries, the proposed system uses a deterministic routing mechanism to provide structured and controlled response to medical queries. The routing strategy is a decision layer that is placed in between user input processing and response generation. Its main role is to categorize queries received into functional categories that have been defined and call the best suitable module to generate the response.

In the event that a user sends a message through WhatsApp, the query is first preprocessed and an intent is detected. The routing module examines the semantic structure and key patterns that the input has to identify the route of the correct response. The queries are divided into four great domains: analysis of symptoms, drug-information search, hospital-discovery and general-medical-knowledge. Depending on the intent that has been identified, the system instead of invoking the language model on each request activates the special tool module that corresponds to the identified intent.

As an example, matches on queries about symptoms lead to the activation of the Symptom Analysis Tool that takes in the input query and gives a structured response including a list of potential causes, a list of general remedies, and a list of advisory warnings. Questions concerning drugs trigger the Drug Intelligence Tool which retrieves information including uses, precautions and side effects. Location-based queries initiate the Hospital Finder Tool using the Google Places API, whereas the general medical awareness queries are sent to the RAG Engine to generate document-grounded responses.

This deterministic routing plan lowers the unnecessary generative processing and its reliability is improved by applying task-specific workflows. It is also better at making the system comprehensible, where one can trace each response to a particular module as opposed to an

individual monolithic AI system. Also, routing eliminates any overlap of functional modules and domain separation that is highly essential in safety-sensitive healthcare applications.

Through integrating deterministic intent-based routing with retrieval-augmented generation, the proposed chatbot will be able to balance the structured decision-making with intelligent contextual understanding.

3.6 Safety and Emergency Detection Mechanism

As the healthcare field is a safety-important sphere, the proposed AI-Driven Public health medical assistant includes a specific safety and emergency alert system that helps to avoid harmful or erroneous results. Although the RAG framework provides the factual basis, an extra safety measure is required to address the situations with the high risk of medical conditions and implement the use restrictions imposed by the ethical considerations.

The safety system works on two levels, which are pre-response risk detection and post-response validation. At the pre-response stage, the incoming user queries are recognized as having high-risk keywords and patterns related to an emergency condition like chest pain, difficulty breathing, stroke symptoms, excessive bleeding, or unconsciousness. In case some risk indicators are identified, the usual routing process is canceled and the system is focused on the immediate instructions. The chatbot does not give detailed informational answers but rather tells the user to consult with a professional medical help immediately and opens the hospital finder module where relevant.

Post-response validation step is done to confirm that the outputs generated meet the non-diagnostic predefined constraints. The system filters the replies to avoid prescription recommendations, definite diagnosis, or treatment that can be taken by the clinicians. The language templates are designed in such a way that they will deliver educational information, including the possible reasons and overall preventive practices, and it will constantly guide the reader to seek the help of certified medical workers. This filtering mechanism is used based on constraint and helps in improving ethical compliance and minimizing risks of liability.

Also, the fallback response mechanism is provided to ambiguous queries or unsupported queries. When the system is unable to identify or recall pertinent information with a high level of confidence, then it puts in place a neutral warning message instead of producing speculative

information. Such a conservative response approach minimizes the risk of hallucinations even more and strengthens the responsible use of AI.

The safety layer, which combines emergency prioritization, output filtering, and fallback, makes sure that the chatbot works within clearly defined boundaries that may be used in the public health awareness setting. This systematic safety implementation is required to facilitate the reliability, trust, and ethical adherence of the real world healthcare communication setting.

3.7 Integration and Deployment Approach

The last step of the methodology is the integration of all of the functional modules and the implementation of the AI-driven system into a real-time communication setting. The integration strategy was in such a way that it coordinated the modularity of the backend processing elements, the system of retrieving vectors, routing logic, safety mechanism, and external service APIs. The system structure is a layered structure with each module functioning independently but interacts with the others via a clearly defined interface.

FastAPI was used to develop the backend framework and handles the HTTP requests, management of tools and response formatting. FastAPI also supports asynchronous request processing, which allows processing user requests efficiently in real-time. RAG engine, routing module and safety validation components work in the back-end environment and by ensuring that there is controlled processing of any response before any response is sent to the user.

To deploy the chatbot, the Twilio WhatsApp API was interconnected with the chatbot to support real-time communication on the WhatsApp messaging platform. The webhook endpoints that are set up in the FastAPI application accept incoming user messages. After going through the retrieval, routing and safety layers, the resultant response is sent back to the user through the twilio messaging interface. The communication system is a two-way process that is based on webhook and makes the communication between the chatbot and end users seamless.

Moreover, the Hospital Finder Tool was also incorporated with the help of the Google Places API that helps to find the information about healthcare facilities by their location. Upon a user providing location information or asking to know the nearest hospitals, the system will query the Places API and provide structured data with the name of the hospital, address, and rating.

This is the integration that takes the system further to informational assistance to actionable assistance.

The first deployment was done in secure tunneling services on a confined development setting and later moved to stable execution. During the integration, the response latency, API reliability and error handling mechanism were observed to facilitate a smooth end-to-end interaction. The completed implemented system is fully functional, including the ability to receive messages sent by the user and provide grounded medical response through WhatsApp.

3.8 Testing and Verification Strategy

An organized testing and verification plan was implemented to make sure that the AI-Driven Public Health Medical Assistant would satisfy the functional, performance, and safety specifications before its complete implementation. The system is a combination of various modular modules such as retrieval pipelines, deterministic routing, safety validation, outside API services, hence unit-level as well as system-level testing methodologies were adopted.

Each functional module was individually unit tested to ensure that it was correct then integrated with the rest. The Symptom Analysis Tool was also tested in terms of structured response formatting and correct intent treatment. The Drug Intelligence Tool was confirmed to have full information contents under the fields of usage, side effects, and safety precautions. The retrieval engine based on FAISS was tested by making benchmark queries and ensuring that the displayed document chunks were semantically relevant. The routing mechanism was evaluated on proper intent classification with various query formats and emergency detection module was tested with pre-defined high-risk keywords to ensure that there is proper override behavior.

After the unit validation, integration testing was carried out to determine the end to end system functionality. This step confirmed the full interaction pipeline, starting with the reception of WhatsApp messages via a webhook and the processing of the message, which is then routed to an invocation and executed in response to the safety control, and the transmission of responses. Particular care was taken in making sure that the module invocation was correct, to make sure that symptom queries did not mistakenly invoke drug tools and vice versa. Response latency and API reliability were also monitored to guarantee that the real time performance was acceptable.

Checking There was checking to determine that the system implementation met the previously defined functional requirements, such as deterministic routing, document-grounded responses, safety filtering, and real-time deployment. Simulated real world user scenarios were used in the validation with the help of various types of queries, emergency situations, ambiguous inputs, and location based requests. These testing procedures had the effect of making sure that the chatbot acted in a similar way, that it had followed the safety limits and that it could perform consistently in the circumstances of real use.

The correctness, reliability and responsible AI conduct of the system were systematically developed through unit testing, integration testing, verification and validation before the system was put into operation.

Chapter 4
Project Management

The public health medical assistant developed was significantly successful because of the project management process. The implementation of the project was in a planned and phases approach where planning, development, integration, testing and release phases were included. All stages were set in accordance with specific milestones and deliverables in order to make succession. The Gantt chart (Fig 4.1) is used to visually represent the workflow timeline, whereas this section explains how the tasks will be carried out in the development lifecycle, on a chronological basis.

4.1 Project timeline

The project was implemented within a span of about four months which was between January and April following the academic schedule. The construction was done in an iterative and milestone-based process using Agile and allowed the process of continuous improvement and refinement of the system components.

The major Milestones and Activities

Month 1 -January: Problem Definitions and Research

- Finding out the societal health issues in rural and semi-urban settings.
- The scope and definition of project objectives.
- Literature review of healthcare chatbots, LLMs, and RAG systems.
- Feasibility analysis of AI-based medical assistants.
- Selection of core technologies (RAG, FAISS, FastAPI, WhatsApp API)

Month 2 – February: Knowledge Base Preparation

- Collection of medical documents and datasets
- PDF extraction and preprocessing.
- Text cleaning and normalization: Semantics and syntax.
- Adoption of text chunking strategy.
- First generation of embedding and testing.

Month 3 – March: Development of the core system

- FAISS vector database was developed.
- Implementation of Retrieval-Augmented Generation (RAG) pipeline
- Integration of embedding models and semantic retrieval
- Initial experimentation of document-grounded response generation.
- Symptom, Drug, Hospital Finder Modular tools were developed.

Month 4 – April: Integration, Deployment, and Testing

- Consecutive application of deterministic routing mechanism.
- Integration of safety and emergency detection module
- Backend development FastAPI.
- WhatsApp integration with the use of Twilio API.
- Testing of hospital finder using Google Places API
- End-to-end system validation and performance testing
- Bug fixing and documentation: Final documentation.

4.1.1 Visualization of Timeline

Fig 4.1 Gantt Chart

The Gantt chart of the project timeline is presented in Fig. 4.1, and it identifies the duration and sequences of the activities involved in the project and milestones accomplished. It gives an effective visual representation of the way the various stages of development were planned and

implemented throughout the years that made the project management efficient and the achievement of the goals on time.

Table 4.1 Project Planning Timeline

Phase	**Task**	**Duration**	**Timeline**	**Milestone**
Phase 1	Problem Identification	1 Week	Week 1	Topic finalized
Phase 2	Literature Review	2 Weeks	Week 2–3	Research completed
Phase 3	Requirement Analysis	1 Week	Week 4	Functional requirements defined
Phase 4	System Design Planning	1 Week	Week 5	Architecture finalized
Phase 5	Tool & Technology Selection	1 Week	Week 6	Tech stack finalized

In table 4.1, the planning phase timeline is shown and it is in this table that the underlining aspects of the project were made. Topics approval, completion of the literature survey and methodology design are the major milestones that were attained during this period.The organized methodology has helped make sure that the issue was well established, the right technologies like RAG, FAISS, and FastAPI were chosen, and system architecture was created prior to the implementation. This step was crucial in minimizing the development uncertainties and making sure that it was in tandem with project objectives.

Table 4.2 Project Implementation Timeline

Phase	**Task**	**Duration**	**Timeline**	**Milestone**
Phase 6	Data Collection & Preprocessing	2 Weeks	Week 7–8	Knowledge base prepared
Phase 7	Vectorstore Creation (FAISS)	1 Week	Week 9	Embedding & indexing completed
Phase 8	RAG System Development	2 Weeks	Week 10–11	Context-based response generation
Phase 9	Tool Development (Symptom, Drug, Hospital)	2 Weeks	Week 12–13	Modular tools completed
Phase 10	Routing & Safety Implementation	1 Week	Week 14	Intelligent routing enabled

Phase 11	WhatsApp Integration (Twilio)	1 Week	Week 15	Live chatbot deployment
Phase 12	Testing & Validation	1 Week	Week 16	System verified

The implementation stage was associated with the chatbot system development, such as knowledge base creation, the RAG pipeline creation, routing logic, safety measures, WhatsApp integration, and testing. Table 4.2 shows the implementation process, which was done step by step in the development of the system. The process of work started with the preparation of knowledge base and went on to integrate RAG, develop and deploy the tools. The fact that the special testing period was implemented guaranteed reliability of the system and validation of its performance. Provided that the phased development is implemented based on the Agile methodology, it is possible to make improvements and reveal the problems early.

4.2 Risk Analysis (PESTLE Analysis)

The PESTLE analysis was done as a way of assessing the external environments that are likely to influence the success of the project. This analysis will take into consideration Political, Economic, Social, Technological, Legal and Environmental factors which will allow taking the risk proactively and plan the strategy.

Table 4.3 PESTLE Analysis for the Proposed System

Factor	**Risk**	**Impact**	**Mitigation Strategy**
Political	Changes in government health policies	May affect data integration	Design system to be adaptable to multiple data sources
Economic	Limited funding/resources	Affects scalability	Use open-source tools (FAISS, FastAPI)
Social	Lack of digital literacy in rural areas	Reduced usability	Use WhatsApp (familiar platform)
Technological	API failures (Twilio/Google)	Service disruption	Implement fallback mechanisms
Legal	Data privacy concerns	Compliance risks	Avoid storing personal health data
Environmental	Network/internet issues	Accessibility problems	Optimize system for low bandwidth

Table 4.4 Project Risk Matrix

Risk	**Probability**	**Impact**	**Risk Level**	**Mitigation**
LLM hallucination	Medium	High	High	Use RAG + controlled prompts
Incorrect medical response	Low	High	High	Safety layer + disclaimers
API downtime	Medium	Medium	Medium	Retry + fallback logic
Routing misclassification	Medium	Medium	Medium	Improve keyword detection
Data quality issues	Low	High	Medium	Use verified medical sources

The analysis of the risks reveals that the challenges that present themselves in the AI models are hallucination, reliance on the external APIs, and health safety issues. The mitigation of these risks was done by way of Retrieval-Augmented Generation, deterministic routing, and safety validation mechanisms. The deployment platform WhatsApp also decreases the risk of accessibility, whereas the economic constraints are minimized with the use of open-source tools.

4.3 Project Budget

Budgeting is an essential element of the project management, which makes use of resources the most effective and efficient in the lifecycle of the project development. In the case of the AI-Driven Public Health Medical Assistant, the cost-reduction and efficiency maximization budgeting strategy was based on the use of open-source technologies and the free-tier services. Budgeting process was organized in such a way that it recognized the requirements of resources, estimated the effort of the process and made it feasible throughout the development process including the preparation of the knowledge base, deployment and testing.

Step 1: Identifying the List of all the tasks and resource requirements.

The project was broken down into significant functional phases and these were preparation of knowledge base, development of vectors store, RAG, development of tools, routing and safety testing, deployment of WhatsApp, testing and documentation. The resources needed to carry out each stage that included software tools, APIs, computing infrastructure, and development time were established.

Step 2: Team Availability

The project occurred as a result of the collaboration between a group of contributors working throughout the academic course. The allocation of tasks was to be done according to the complexity of modules in order to balance the workload on the various phases which included the backend development, the design of the AI pipeline, and integration of the system. The highest points of development like the implementation of RAG and WhatsApp were synchronized with the availability of the maximum number of team members.

Step 3: Estimation of Duration of tasks

The time estimation of each task was pegged on:

- Complexity of implementation (e.g., RAG pipeline, routing logic)
- Dependency between modules (e.g., vectorstore before retrieval system)
- Required testing and debugging cycles

These estimates were in line with the project schedule that was outlined in Section 4.1.

Step 4: Application of Past experience and information

The development effort estimation was with the help of the previous experience in the development of AI based systems and software development. No expensive training stages were required because the system takes the form of a pre-trained language model (MedGemma). This greatly lowered the infrastructure needs unlike in the conventional machine learning projects.

Step 5: Budget Estimation

Cost-efficiency of the project was considered, i.e. the use of the open-source tools and free-tier services. The costs considerations made were as follows:

- Use of open-source frameworks such as FastAPI, LangChain, and FAISS
- Use of pre-trained embedding models and local LLM (MedGemma)

- Minimal reliance on paid APIs (Twilio and Google Places within free limits)
- As Local deployment environment Development and testing environment.

Table 4.5 Project Budget Estimation

Resource	**Description**	**Estimated Cost (INR)**
Development Tools	FastAPI, LangChain, FAISS (Open-source)	0
LLM Model	MedGemma (Local deployment)	0
Twilio WhatsApp API	Messaging service (trial/free tier)	0–2000
Google Places API	Hospital finder (limited usage)	0–3000
Hosting / Deployment	Local / optional cloud usage	0–2000
Miscellaneous	Internet and testing resources	1000

Total Estimated Cost: ₹1000 – ₹8000

Step 6: Budget Monitoring and Resource Utilization

Simple monitoring tools were used to track the utilization and development of the budget. Such parameters as the estimated and actual effort, the API use and the system performance were monitored during development. This allowed transparency and reallocation of effort at critical points like debugging and integration to be efficient.

4.4 Resource Management

4.4.1 Resource Allocation

1. Human Resources

- Project developers (team members)
- Faculty guide and evaluators

2. Software Resources

- FastAPI, LangChain, FAISS
- HuggingFace embedding models
- Twilio WhatsApp API

- Google Places API
- GitHub for version control

3. Computing Resources

- Standard CPU-based laptops
- Local development environments (VS Code, Python)
- Local storage and vector database

4.4.2 Project Monitoring and Tracking

The following was used to continually monitor the project progress:

- Agile based progress reviews on a weekly basis.
- Version control using GitHub commits
- Task coordination by means of mutual communication platforms.
- Documentation tracking using collaborative tools
- Periodic milestone assessment by faculty.

This systematic form of monitoring allowed the project goals to be met on time and coordination among the team members was done efficiently.

Chapter 5
Analysis and Design

System development requires analysis and design, which identify the structure, behaviour and the implementation plan of a solution. The analysis stage is aimed at determining system requirements, the needs of the user, and non-functional and functional specifications. The design stage converts these requirements to an organized architecture, which specifies the way the various constituents interrelate to bring forth the intended functionality. In the proposed AI-Driven Public Health Medical Assistant, a design was done to develop a modular, scalable and safe AI-driven architecture that incorporated Retrieval-Augmented Generation (RAG), deterministic routing, and real-time messaging deployment based on analysis to identify the challenges in healthcare communication and requirements of the system. It is a systematic strategy that makes the system reliable, efficient, and applicable in the real-life healthcare awareness application.

5.1 Requirements

The users of the AI-Driven Public Health Medical Assistant identified its requirements as they are connected to the necessity to offer the users convenient, reliable, and safe healthcare information in the form of conversation. The system requirements consist of functional requirement, data processing requirement, system management, security requirement and deployment requirement.

Table 5.1 Summarizing Requirements

Category	**Description**
Purpose	To develop an AI-powered chatbot that provides disease awareness, symptom guidance, drug information, and hospital location services through WhatsApp

Behaviour	The system should process user queries and route them to appropriate modules (symptom, drug, hospital, or general knowledge) and generate safe, structured responses
System Management	The system should manage query processing, routing, retrieval, and response generation efficiently in real-time
Data Analysis	The system should perform semantic similarity search using vector embeddings for accurate information retrieval
Application Deployment	The chatbot should be deployed via WhatsApp using Twilio API and accessible in real-time
Security	The system should avoid storing sensitive user data and enforce safe, non-diagnostic responses
User Interface	The system should provide a simple conversational interface through WhatsApp for ease of use

5.1.1 System Software Requirement Phase

1. Initial Conditions

- User sends a message via WhatsApp
- Backend system is active and connected

2. Input Parameters

- User query (text)
- Optional location data

3. System Outcomes

- Context-aware response
- Hospital suggestions (if required)
- Safety advisory (if emergency detected)

4. System Constraints

- No medical diagnosis or prescriptions
- Limited to informational responses
- Dependent on API availability

5.1.2 Data Requirements

1. Data

- Medical documents for knowledge base
- Query input data
- Location data (optional for hospital finder)

2.System Management Requirements

- Efficient routing of queries
- Modular tool integration
- Real-time response handling

3. Security Requirements

- No storage of personal health data
- Emergency detection and override
- Safe response filtering

4. User Interface Requirements

- WhatsApp-based interface
- Simple text-based interaction
- Location-sharing capability

5.2 Block Diagram

The AI-Driven Public Health Medical Assistant functional block diagram is a general structure of the system and communication between the various modules. The system is modular,

meaning that every block carries out a particular task in order to guarantee an efficient processing of the user queries and production of safe and contextual responses.

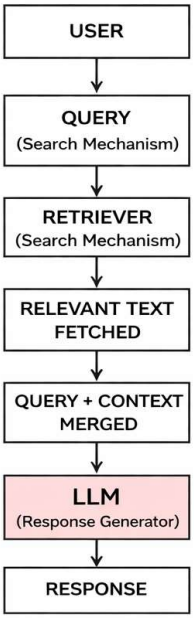


Fig 5.1 Functional Block Diagram

The user interface starts with the WhatsApp interface, which is the major input medium in the system. The user messages are also received on the Twilio WhatsApp API which serves as a communication channel between the user and the back-end system. The received request is then redirected to the FastApi backend where all the action and coordination is done.

In the backend, the routing module examines the query of the user and identifies the path to process them. Depending on the purpose of the query, it sends this query to one of the functional modules, such as the Symptom Analysis Tool, Drug Information Tool, Hospital Finder Tool or the Retrieval-Augmented Generation (RAG) engine. The RAG module communicates with the FAISS vector database to find a relevant medical information by the semantic similarity search.

The retrieved or processed information is then subject to the safety layer which is used to make sure that the response is constrained to a set of pre-defined constraints including non-diagnostic output, emergency response and ethical response. The response is finally processed and validated and sent back to the user via the WhatsApp interface completing the interaction cycle.

A block diagram is effective to show the logical flow of data and control in the system in such a way that the responsibility of communication, processing, intelligence, and safety layers is

clearly divided. The modular architecture is also more scalable, maintainable, and reliable and, therefore, it is appropriate in the real world healthcare awareness applications.

5.3 System Flow Chart

The system flowchart is the chronological implementation of the AI-Driven Public Health Medical Assistant, the way the input of the user is received, analyzed, and transformed into a structured output. It gives a clear visualization of the steps of operation, decision making points as well as interaction of the system components.

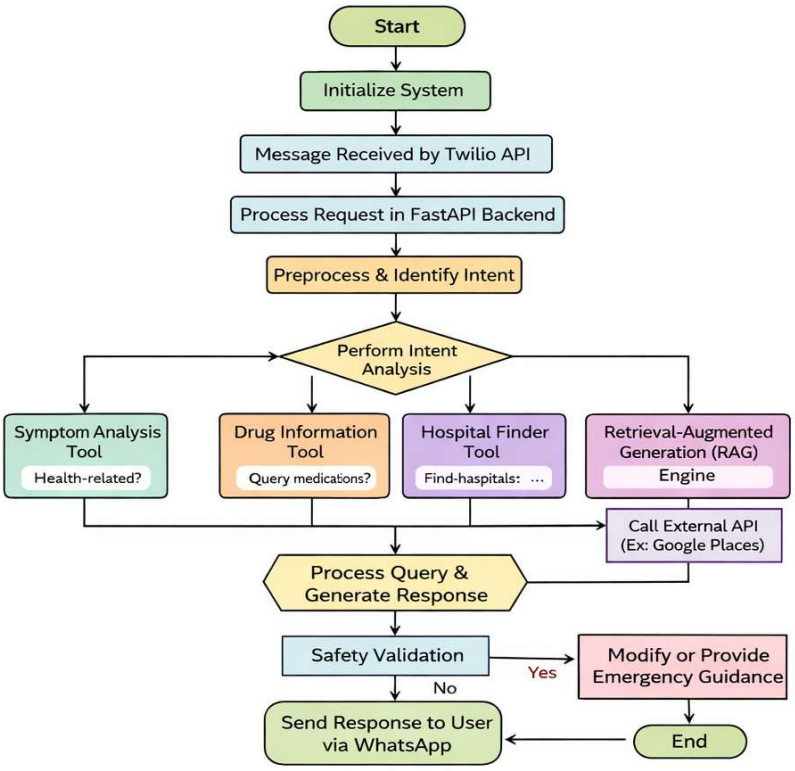


Fig 5.2 System Flow Chart

This flow starts with the initiation of the system and then the user sends a message via the WhatsApp interface. The Twilio API receives the message and routes it to FastAPI backend server to process the message. When the request is received, the system preprocesses the input and forwards the same to the routing module.

Routing module carries out intent analysis in order to identify the nature of query. According to this rule, the query would be sent to either of the functional modules including the Symptom Analysis Tool, Drug Information Tool, Hospital Finder Tool, or the Retrieval-Augmented

Generation (RAG) engine. In case of RAG path being chosen, information is obtained in the FAISS vectors database. Google places are included as external APIs to get the location-based data in the case of hospital queries.

Once the query is processed the output that is generated is subjected to the safety validation layer. This would make sure that the response is within a set of constraints assigned like no medical diagnosis, identification of emergencies and ethical compliance. In case any risk is detected, the system adjusts the response by doing so or prioritising emergency guidance.

And the last step is to send the validated response to the user via the WhatsApp interface and that completes the interaction cycle. The flowchart is very useful to illustrate the rational order of the operations and emphasize on the combination of routing, retrieval, safety, and communication modules, which makes the system strong and suitable to the real-time healthcare support.

5.4 Standards

Standards are necessary to guarantee the interoperability of the system, security, reliability and scalability. The proposed AI-Driven Public Health Medical Assistant is taking into account the following standards:

1. Communication Standards

- HTTP/HTTPS protocols are used for communication between:
    - WhatsApp interface
    - Twilio API
    - FastAPI backend
- HTTPS ensures secure data transmission using TLS (Transport Layer Security)
- RESTful API architecture is followed for:
    - Modular communication
    - Scalability
    - Easy integration

2. Data Format Standards

- JSON (JavaScript Object Notation) is used for data exchange
- Benefits of JSON:
  - Lightweight
  - Human-readable
  - Compatible with web-based systems

3. Security Standards

- TLS encryption for secure communication
- No storage of sensitive user data
- Safety constraints enforced:
  - No diagnosis
  - No prescriptions
- Relevant standards:
  - ISO/IEC 27001 – Information Security Management
  - ISO/IEC 27701 – Data Privacy
- Ethical AI practices followed to prevent misuse

4. AI and System Architecture Standards

- ISO/IEC 30141 – IoT Reference Architecture (layered design concept)
- ISO/IEC 42001 – AI Management System
- Supports:
  - Modular system design
  - Interoperability
  - Scalable architecture

5. Benefits of Using Standards

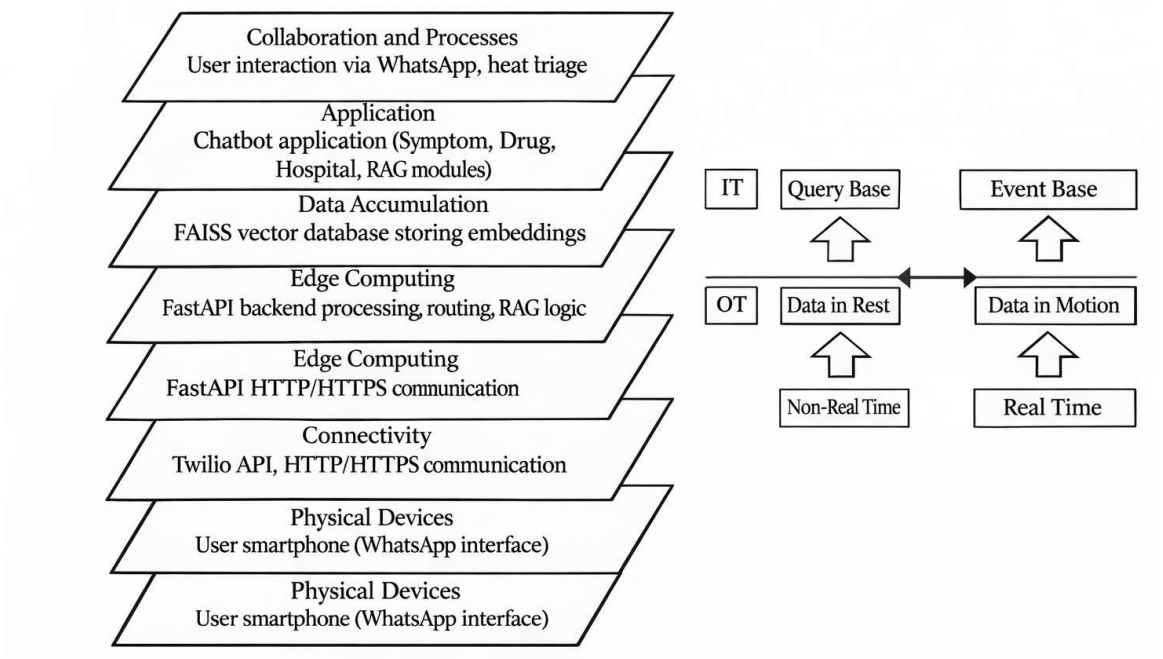
- Better interoperability in the system parts.

- Improved security and privacy of data.
- Reduced development complexity
- Enhanced reliability and scalability of the system.
- Allows the connectivity with external APIs and services.

5.5 Mapping with IoT World Forum Reference Model

The IoT World Forum (IoTWF) Reference Model offers a seven-layer architecture in a structured manner to develop scalable and interoperable systems. Though the proposed system is software-based, it can be traced to the model of the IoTWF using the interaction between devices used by users, layers of communication, data processing, and application services.

Fig 5.3 Mapping with IoT World Forum Model



The IoTWF mapping gives a stratified view of the way the system works at the point of user interaction to the processing at the back end. In the bottom, the system communicates with the user devices in the form of smartphones, and the connectivity layer provides a secure connection by using APIs and internet protocols. The edge computing layer is used to perform core processing, such as routing, retrieving and response generation.

The data accumulation and abstraction layers deal with the storage and transformation of medical information through the use of vector databases and structured outputs. The top layer

deals with an interaction and health awareness outcomes of the chatbot, whereas the application layer deals with the functional modules of a chatbot.

The security is implemented on all levels, such as secured communication, safe generation of responses, and prevention of storing sensitive data. This stratified mapping shows that the system is based on a scalable and modular architecture in accordance with the principles of IoT design.

Table 5.4 Mapping Project Layers with IoTWF Reference Model

Layer	IoTWF Layer Description	Project Layer Mapping	Security Consideration
7	Collaboration & Processes	User interaction via WhatsApp, healthcare awareness usage	Safe usage guidelines, no misuse of medical info
6	Application	Chatbot application (Symptom, Drug, Hospital, RAG modules)	Response validation, no diagnosis
5	Data Abstraction	Processed responses, structured outputs	Controlled data formatting
4	Data Accumulation	FAISS vector database storing embeddings	Secure storage, no sensitive data
3	Edge Computing	FastAPI backend processing, routing, RAG logic	Input validation, error handling
2	Connectivity	Twilio API, HTTP/HTTPS communication	TLS encryption
1	Physical Devices	User smartphone (WhatsApp interface)	Device-level security

5.6 Domain Model Specification

The domain model is the definition of major entities, components, and relationships of the AI-Driven Public Health Medical Assistant system. It gives abstracted description of the interaction of various elements regardless of the particular technologies or implementation arrangements. The domain model assists in the interpretation of how user queries are turned

into significant responses by the interaction of physical entities, virtual representations, services, and system resources.

Table 5.5 Description of Domain Model

Component	**Description**
Physical Entity	Users seeking healthcare information through WhatsApp
Virtual Entity	Digital representation of user queries and medical information processed by the system
Device	User smartphone and communication interface (WhatsApp)
Resource	Backend services such as FAISS vector database, APIs, and AI models
Service	Chatbot services including symptom analysis, drug information, hospital finder, and RAG engine

The user who is communicating with the chatbot to get healthcare information is the physical entity in the proposed system. All user requests are transformed into virtual objects, which are the digital version of the request that is acted upon by the system. The interaction device is a smartphone that runs WhatsApp that serves as an interface between the user and the backend system.

The system is based on several resources, such as semantic retrieval relying on FAISS vector database, query processing relying on pre-trained embedding models, and third-party APIs, such as Twilio and Google Places, to communicate and provide location services. These resources facilitate the system to process, retrieve and generate correct responses.

The services layer comprises of modular chatbot services including symptom analysis, drug information search, hospital search and general medical knowledge generation based on the RAG framework. These services interrelate with the system resources and generate meaningful outputs based on user queries.

The domain model emphasizes the relationship between user input, system processing, and service delivery and is an excellent abstraction of the system workings. Such a formal representation provides modularity, scalability, and effective integration of various components of the chatbot architecture.

5.7 Communication Model

The communication model identifies the way various elements of the system communicate with each other and share information. According to the needs of the AI-Driven Public Health Medical Assistant, the most appropriate communication model is the Request-Response model. The model is common in web applications and API-based systems whereby a client makes a request and a response is provided to the client.

Selected Communication Model: Request–Response

Communication in the proposed system is of request response based where the user initiates a communication pattern and the system proceeds to process it and respond.

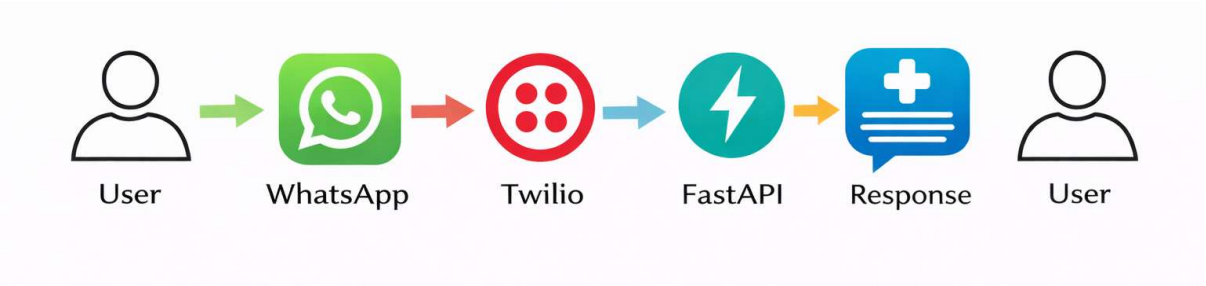


Fig 5.4 Request–Response Communication Model

Working of the Communication Model:

- The user sends a query through the WhatsApp interface
- The message is received by the Twilio API
- Twilio forwards the request to the FastAPI backend via webhook
- The backend processes the request using:

  - Routing module
  - RAG engine
  - Tool modules (Symptom, Drug, Hospital Finder)
- The system generates a response
- The response is sent back to the user via WhatsApp

The request–response model is suitable for this system because:

- It supports real-time interaction between user and chatbot
- It aligns with REST API architecture used in FastAPI
- It ensures simplicity and reliability in communication
- It allows easy integration with external services like Twilio and Google APIs
- It supports scalable deployment for multiple users

Advantages

- Simple and easy to implement
- Efficient for real-time query processing
- Widely supported across web and messaging platforms
- Enables modular backend design

5.8 IoT Deployment Level

The level of IoT deployment determines the way an application is designed in regard to the interaction between devices, data processing, communication, and cloud integration. The system can be best understood as being aligned with IoT Deployment Level 3, based on the architecture and functionality of the AI-Driven Public Health Medical Assistant.

Selected Deployment Level: IoT Level 3

IoT Level 3 consists of:

- Devices connected to a cloud-based or centralized processing system
- Data processing performed at the backend/server level
- Communication through internet-based protocols
- User interaction through applications or interfaces

Mapping to the Proposed System

- Devices: User smartphones using WhatsApp
- Connectivity: Internet-based communication via Twilio API (HTTP/HTTPS)
- Processing: Backend processing using FastAPI, routing logic, and RAG framework
- Data Storage: FAISS vector database storing embeddings
- Application Layer: Chatbot services (Symptom, Drug, Hospital Finder, RAG)

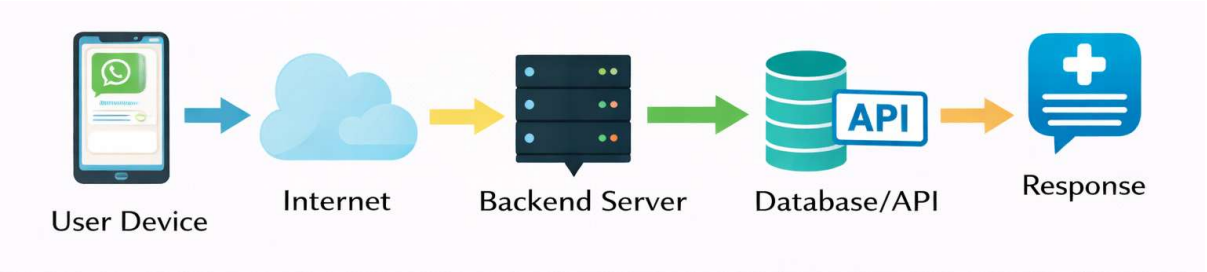


Fig 5.5 IoT Deployment Level 3 for AI Medical Chatbot

Working of Deployment Level:

1. The user interacts with the system using a smartphone
2. The query is transmitted via the internet using WhatsApp and Twilio API
3. The backend server processes the request using AI models and retrieval systems
4. Relevant data is fetched from vector databases or APIs
5. The processed response is sent back to the user

IoT Level 3 is suitable because:

- It supports centralized processing (important for AI models)
- It allows integration with cloud/APIs (Twilio, Google Places)

- It enables real-time communication with users
- It supports scalability for multiple users
- It avoids dependency on physical sensors or edge devices

5.9 Functional View

The functional view depicts the system in a grouping of its operations into major functional parts. It allows one to comprehend the interaction of various components of the system to handle user queries and produce answers.

The suggested AI-Based Public Health Medical Assistant may be subdivided into the following functional groups:

- **Device Layer:** User smartphone interacting through WhatsApp
- **Communication Layer:** Twilio API handling message transmission
- **Processing Layer:** FastAPI backend managing routing, RAG, and tool execution
- **Service Layer:** Functional modules such as Symptom Tool, Drug Tool, Hospital Finder, and RAG engine
- **Data Layer:** FAISS vector database and external APIs (Google Places)
- **Security Layer:** Safety validation and emergency detection mechanisms
- **Application Layer:** Final chatbot interface delivering responses to users
-

5.10 Mapping Deployment Level with Functional View

This section is the mapping of the level of deployment of the IoT (Level 3) and the functional components of the system. It demonstrates the correspondence between each functional layer and various levels of system functioning and communication.

Table 5.6 Mapping Deployment Level with Functional View

IoT Deployment Level Component	Functional View Component
Device	User smartphone (WhatsApp)
Connectivity	Twilio API, HTTP/HTTPS
Edge/Processing	FastAPI backend, routing module
Data Storage	FAISS vector database
Services	Symptom, Drug, Hospital, RAG modules
Security	Safety layer, emergency detection
Application	Chatbot interface (WhatsApp responses)

1. Device layer corresponds to the user smartphone

2. Communication layer maps to internet-based APIs (Twilio)

3. Processing layer aligns with backend computation (FastAPI, routing, RAG)

4. Service layer represents chatbot functionalities (Symptom, Drug, Hospital Finder)

5. Data layer corresponds to storage and retrieval (FAISS, APIs)

6. Security layer ensures safe operation across all levels

7. Application layer delivers responses to the user

5.11 Operational View

The operational view determines the deployment and operational appearance of the system in the real-world setting, as it relates to communication alternative, service hosting, storage, and execution of applications. It offers a real-life insight into the way the AI-Driven Public Health Medical Assistant works in the course of the run.

1. Communication Options

The system operates via the internet-based communication with the HTTP/ HTTPS protocols. WhatsApp is used to send user queries and the Twilio API to process them, sending them to the FastAPI backend via webhooks. This facilitates a two way (bidirectional) real time communication between the user and the system.

2. Service Hosting Options

Fast API is used to host the backend services and is in charge of processing requests, routing and producing responses. The system can also be scaled to the cloud to support large scale use, but is made to support local deployment during development.

This modular architecture enables independent services like routing, RAG processing and safety validation to run in the backend environment.

3. Storage Options

It is a system that embeds medical documents in a FAISS-based vector database. This enables effective semantic access of pertinent information. Besides the use of vector storage, the external APIs (e.g., Google Places) are also utilized to retrieve real-time information (e.g., nearby hospitals). The system does not store sensitive user data, which makes it compliant and private.

4. Device Options

Its main user gadget will be a smartphone with WhatsApp installed, which will provide easy access to users in the rural and semi-urban regions. The system is not dependent on special hardware, which makes it economical and scalable. Standard computing devices are used to do the back end processing that can run python based services .

5. Application Hosting Options

The chatbot application is deployed in FastAPI server with the WhatsApp platform via Twilio. Ngrok is a tool that is used during development to open local servers to the internet. The system can be implemented on cloud production systems, which can be highly available and scaled.

Suitability:

- Real-time communication and response generation
- Scalability for multiple users
- Cost-effective deployment using open-source tools
- High accessibility through widely used platforms like WhatsApp
- Secure and privacy-aware operation

5.12 Other Design Aspects

This part of the specifications addresses other design requirements, which contribute to the overall functionality and effectiveness of the system.

1. Process Specification

The system will have a systematic approach towards responding to user queries:

1. User sends a message via WhatsApp
2. Message is received through Twilio API
3. FastAPI backend processes the request
4. Routing module determines query type
5. Appropriate tool or RAG system is invoked
6. Response is generated and validated
7. Final output is sent back to the user

This process ensures efficient handling of queries with minimal latency and high reliability.

2. Information Model Specification

The information model defines how data is structured and processed within the system:

- Input: User queries (text and optional location data)

- Intermediate Data:
  - Query embeddings
  - Retrieved document chunks
  - API responses
- Output:
  - Structured and safe medical response

The accuracy of retrieval and relevancy of responses is enhanced by the use of semantic interpretations of user queries made possible by the use of the vector embeddings.

3. Service Specification

The system provides multiple services through modular components:

- Symptom Analysis Service: Provides possible causes and general advice
- Drug Information Service: Provides usage, side effects, and precautions
- Hospital Finder Service: Provides nearby hospital details using location
- RAG-Based Knowledge Service: Provides general medical awareness information

Every service is autonomous but coordinated by the routing mechanism, which provides modularity and scalability.

4. Design Considerations

- Modular architecture for easy scalability
- Safety-first approach to avoid harmful responses
- Use of RAG to reduce hallucination
- Real-time deployment via WhatsApp for accessibility
- Minimal resource usage for cost efficiency

Chapter 6

Hardware, Software and Simulation

6.1 Hardware

The suggested AI-Driven Public Health Medical Assistant is mainly a software-defined implementation and it will not entail the use of any special hardware materials like sensors, microcontrollers, and embedded systems. Nonetheless, the development, testing and deployment were done using standard computing hardware. The system was created and implemented on the general-purpose computing systems e.g. laptops and desktops with enough computing power to execute Python based backend services and AI models. The system did not need any special hardware as it uses built-in models and effective retrieval-based processing systems. This minimizes hardware dependency to a great extent as well as makes the system affordable and available.

The main hardware materials are:

- User Interface: Smartphone with WhatsApp on it to interact with the user.
- Development System: Coding, testing and deployment Laptop or desktop.
- Network Infrastructure: Internet access to be used in API communication and real-time messaging.

This is because the system can be deployed in resource-constrained environments like rural and semi-urban areas due to the lack of specialized hardware.

6.2 Software Development Tools

The system was developed using a number of software tools and platforms used to facilitate the development, integration, testing, and deployment.

1. Integrated Development Environment (IDE)

- Visual Studio Code (VS Code) was used for writing and debugging code
- Provides features such as:
    - Syntax highlighting
    - Debugging tools
    - Extension support

2. Programming Language

- Python was used as the primary programming language
- Chosen for:
    - Rich AI/ML libraries
    - Fast development
    - Backend compatibility

3. Backend Framework

- FastAPI was used to build the backend server
- Features:
    - High performance
    - Asynchronous request handling
    - REST API support

4. AI and NLP Framework

- LangChain was used for:
  - Orchestrating LLM workflows
  - Managing retrieval pipelines
- HuggingFace models used for:
  - Generating embeddings
  - Semantic processing

5. Vector Database

- FAISS (Facebook AI Similarity Search) used for:
  - Storing embeddings
  - Performing similarity search
- Enables fast retrieval of relevant medical information

6. APIs and External Services

- Twilio WhatsApp API
  - Enables chatbot interaction via WhatsApp
- Google Places API
  - Provides location-based hospital information

7. Version Control

- GitHub
  - Used for code versioning and collaboration

8. Development Utilities

- ngrok
  - Used for exposing local server to the internet
  - Enables webhook integration with Twilio

9. Configuration Procedure (Important for marks)

- Install Python and required libraries
- Set up FastAPI server
- Configure environment variables (.env file)
- Set up FAISS vector database
- Configure Twilio webhook URL using ngrok
- Integrate Google Places API key
- Run backend server and test endpoints

6.3 Software Code

The system is coded in modular python code with each component being designed as a functional unit. Routing logic, RAG processing, tool implementations and API integration are the key modules.

Example: Routing Logic

```
def route_query(query):
    if "hospital" in query.lower():
        return "hospital"
    elif "medicine" in query.lower():
        return "drug"
    elif "pain" in query.lower():
        return "symptom"
    else:
        return "rag"
```

Explanation:

- This function classifies user queries
- Determines which module should handle the request
- Enables deterministic routing

Example: FAISS Retrieval

```
docs = vectorstore.similarity_search(query, k=3)
```

Explanation:

- Retrieves top 3 relevant document chunks
- Uses semantic similarity
- Supports RAG pipeline

Example: WhatsApp Integration

```
@app.post("/whatsapp")
async def whatsapp_reply(request: Request):
    form = await request.form()
    user_msg = form.get("Body")
```

Explanation:

- Receives incoming WhatsApp messages
- Extracts user query
- Sends to backend processing

Its modular nature is able to guarantee maintainability, scaling and simple integration with other functionalities.

6.4 Simulation

The system is software based and hence simulation was done by functional testing and real time interaction as opposed to hardware simulation.

1. Simulation Approach

- API testing using local server
- Query-response validation

- Testing different user scenarios

2. Types of Simulation Performed

- Functional Testing
    - Verifying each module independently
- Integration Testing
    - End-to-end system validation
- User Interaction Simulation
    - Testing chatbot responses through WhatsApp

3. Testing Scenarios

- Symptom queries
- Drug-related queries
- Hospital search queries
- Emergency cases

Outcome

- System successfully handled real-time queries
- Responses were context-aware and safe
- Integration between modules worked correctly

Chapter 7

Evaluation and Results

7.1 Test Points

The end-to-end performance of the AI-Driven Public Health Chatbot was tested by defining test points in various functional modules of the system. As the system is software-based, the test points are associated with processing pipeline stages.

Table 7.1 Test Points and Functional Units

Test Point	Functional Unit	Description	Expected Outcome
TP1	Input Layer	WhatsApp message reception	Query received correctly
TP2	Backend	FastAPI processing	Query parsed and forwarded
TP3	Routing	Query classification	Correct module selected
TP4	RAG Engine	FAISS retrieval	Relevant documents retrieved
TP5	Tools	Symptom/Drug/Hospital modules	Structured output generated
TP6	Safety Layer	Emergency detection	Safe response triggered

TP7	Output	WhatsApp response	Final reply delivered

Test Scenarios

- Symptom queries (fever, cough, dengue, etc.)
- Drug-related queries (paracetamol, ibuprofen)
- Preventive healthcare queries
- Hospital location queries
- Emergency scenarios

7.2 Test Plan

The system was tested using structured test cases covering multiple real-world healthcare scenarios.

Test Cases

- TP1: System must receive WhatsApp queries in real-time
- TP2: Routing must correctly classify queries (~70%+ accuracy)
- TP3: RAG must retrieve relevant medical documents
- TP4: System must generate structured and safe responses
- TP5: Emergency queries must trigger safety mechanism
- TP6: Hospital finder must return nearby results
- TP7: System must handle ambiguous queries

Testing Methods

- Black-box testing (valid/invalid queries)
- Unit testing (modules: RAG, routing, tools)

- Integration testing (end-to-end pipeline)
- System testing (complete chatbot flow)
- Validation testing (real-world usage scenarios)

7.3 Test Results

This system was tested on various healthcare queries in categories like symptoms, drugs, preventive care and hospital discovery. The chatbot operates on Retrieval-Augmented Generation (RAG) scheme, according to which, FAISS retrieves appropriate medical records, and the language model forms grounded responses. This will guarantee a reduced hallucination and factual accuracy.

Table 7.2 Example User Queries and Chatbot Responses

User Query	Retrieved Context Source	Chatbot Response Summary
Symptoms of dengue?	FAISS medical document	Provides symptoms like fever, headache, joint pain
Prevent malaria?	Preventive knowledge base	Suggests mosquito control and hygiene
What is paracetamol?	Drug corpus	Explains usage and precautions
Persistent cough?	Symptom database	Gives possible causes + doctor advice
Find hospital	API tool	Returns nearby hospitals

Functional Performance

The system successfully demonstrated:

- Accurate routing of user queries
- Context-aware responses using RAG
- Retrieval from validated medical sources
- Real-time hospital detection via API
- Safe responses with disclaimers

Table 7.3 Performance Evaluation Metrics

Metric	Value	Description
Routing Accuracy	71%	Correct module selection
Response Relevance	91%	Matches user intent
Safety Compliance	97%	Emergency handling
Average Response Time	3.7 sec	End-to-end latency
Hospital Detection Success	80%	Location-based results

Performance Visualization

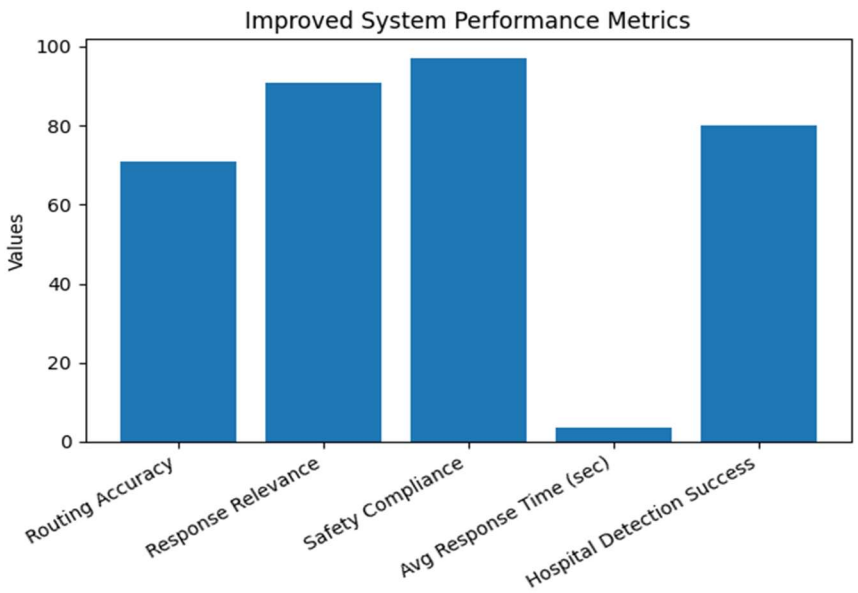


Fig 7.1 Performance Metrics Graph

Fig.7.1 demonstrates that the system has high response relevance (91%) and safety compliance (97%), with efficient response time.

Response Time Analysis

The chatbot takes 2-4 seconds to respond, based on the complexity of the query and the use of API.

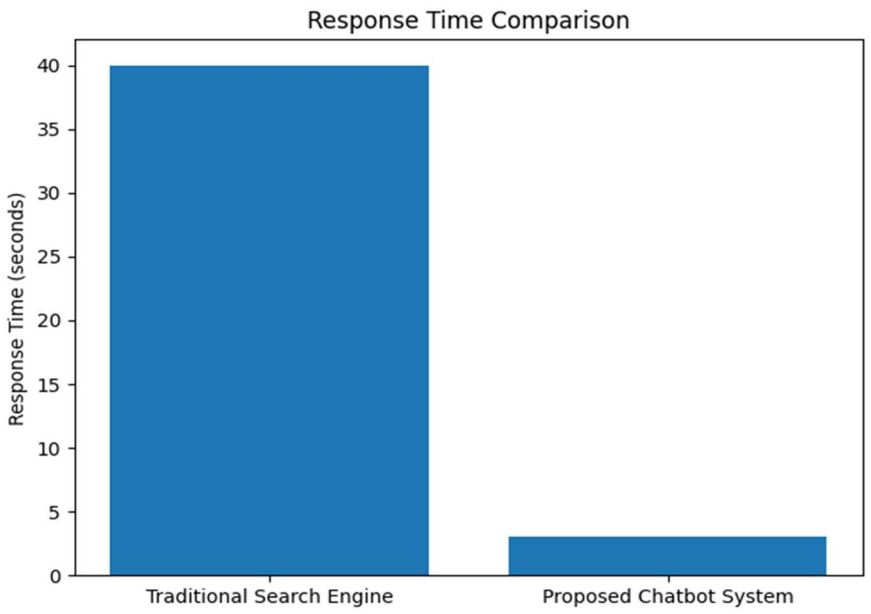


Fig 7.2 Response Time Comparison

Fig. 7.2 , chatbot is much quicker compared to traditional search engines (20-60 seconds), because it does not necessitate a series of search queries, but rather direct answers.

Table 7.4 Comparison with Traditional Search Systems

Evaluation Factor	**Traditional Search Engine**	**Proposed Chatbot System**
Information Access	Multiple sources	Direct answers
Response Structure	Links	Structured response
Ease of Understanding	Technical	Simple
Context Awareness	Low	High (RAG-based)
Risk of Misinterpretation	High	Low
Emergency Handling	Not available	Integrated
Accessibility	Moderate	High (WhatsApp)
User Effort	High	Low
Response Time	20–60 sec	2–4 sec

7.4 Insights

Observations

- RAG significantly improved factual correctness
- Chatbot responses were structured and user-friendly
- Hospital finder produced accurate location-based results
- WhatsApp interface improved accessibility
- System reduced user effort compared to search engines

Limitations

- Not suitable for professional diagnosis
- Routing is rule-based (can be improved)
- Depends on external APIs (Twilio, Google Places)
- No personalization or user history

Overall Evaluation

The system has a high level of performance in terms of offering precise, quick, and context-driven medical information, which is appropriate in the real-world implementation in the field of public health awareness.

Chapter 8

Social, Legal, Ethical, Sustainability and Safety aspects

The creation and implementation of the AI-Driven Public Health Medical Assistant are associated with several aspects that should be considered outside the technical implementation. These are social impact, compliance with laws, ethical responsibility, sustainability and safety. The system communicates with users in a sensitive field like healthcare, hence the need to make sure that it performs in a responsible, secure, and responsible manner that meets expectations in the society. The engineers and developers should make sure that there is a safe, legal, and ethical usage of the system. The use or wrong information may have severe repercussions such as misinformation, delayed medical treatment, loss of trust. Thus, these aspects are very important and should be carefully designed and assessed in order to be deployed successfully in the real world.

8.1 Social Aspects

The chatbot suggested is characterized by a high social value, especially in enhancing accessibility to healthcare by rural and semi-urban communities. The system will give medical information via WhatsApp allowing users to access health guidance without the special applications or technical expertise.

Among these positive effects, there are increased knowledge of diseases, preventive medical care and early detection of symptoms. The system also saves addiction to unreliable web resources and also enables the users to make informed choices regarding seeking medical care. Besides, the fact that a familiar platform such as WhatsApp is used promotes the adoption and use by non-technical users.

There are, however, some difficulties. Digital divide will restrict access to people who lack access to smartphones or internet connection. It is also possible to have the risk of over-reliance of the chatbot by the users, as opposed to visiting the healthcare specialists. In addition, differences in literacy can have an impact on responses interpretation. All these restrictions indicate the necessity to create awareness, educate, and use the system in a responsible way [20].

8.2 Legal Aspects

The legal issues are essential in the implementation of the data protection and healthcare regulations. The chatbot is not used to store personal health data, which is why the risks related to the violation of data privacy are minimized. Nevertheless, the interaction between users and the system still needs to be encrypted with the help of such an encryption protocol as HTTPS.

The system is compliant with the principles of data protection, which are stipulated in the laws like the Digital Personal Data Protection Act (DPDPA), 2023 (India) and the General Data Protection Regulation (GDPR) (EU). These policies focus on the consent of the users, data minimization, and safe data processing. This is because the chatbot does not offer medical diagnosis but only informational guidance thus it does not attract issues of legal liabilities that come with clinical decision-making.

The obstacles are making sure that they comply with various jurisdictions and handling obligations in case of wrong or misinterpreted information. The system contains clear disclaimers and safety warnings to deal with these issues and make sure the use of the system is legal [21].

8.3 Ethical Aspects

Ethics is an important factor in AI-based healthcare system development. Engineers have the major responsibility of making sure that the system is beneficial to the society and would not harm it. The chatbot will not involve diagnosis, prescription, or critical medical decision-making, being rather aimed at the provision of general health information.

Bias is one of the most important ethical issues of the AI systems. In the event that the underlying data is incomplete or skewed, there may be erroneous or misleading responses. In order to solve this, the system employs Retrieval-Augmented Generation strategy to assure that the responses are based on reliable sources of information.

Another major aspect is transparency. Its system makes it clear that it is a system that uses AI and is not a substitute of professional medical advice. Moreover, the chatbot will not cause addictive or misleading communication and will focus on informative and responsible communication.

The principles of ethical AI like fairness, accountability, and transparency are adhered to and the system is made to serve the greater good of the people and avoid causing distrust in the users [22].

8.4 Sustainability Aspects

The system is created keeping in mind sustainability in terms of using the resource, scalability, and environmental impact. As all the chatbot is software-centered and does not involve physical hardware elements, it reduces the use of materials and electronic waste.

This is through lightweight models, local processing and open source tools which provides resource efficiency. FAISS is efficient to use in retrieval, which is less computationally expensive than training of large models. Also, the system can be implemented on the cloud infrastructure, which provides the efficient use of resources and scale according to the demand.

Long term usability is another thing that is aided by the design because the modular architecture will be easy to maintain and update without need to go through the entire system to design it anew. Through decreasing physical healthcare visits to obtain simple information, the system indirectly leads to the decrease in transportation and environmental effects.

In general, the system is in line with the sustainability design principles as it reduces the amount of resources used, maximizes the efficiency, and facilitates the availability of healthcare solutions [23].

8.5 Safety Aspects

The proposed system is sensitive to the issue of safety, particularly because of the medical field that it will be applied. The chatbot has several safety features to avoid dangerous consequences and make it use responsibly.

The system has emergency detection layer that detects high risk queries like chest pain or difficulty in breathing. When this happens, the chatbot takes the safety aspect seriously, hence recommending emergency medical care rather than general information. Moreover, the responses are all filtered to exclude any diagnostic or prescriptive information.

Security in the form of cybersecurity is also taken into account, and secure communication protocols are applied to secure data transfer. The system does not store sensitive user data and this mitigates the threat of data breach. Frequent testing and validation would make sure that the system acts reliably in various situations.

The system guarantees a secure and reliable interaction between users who want healthcare information because of integrating safeguard measures, ethical limitations, and safe communication practices [24].

Chapter 9

Conclusion

The AI-Driven Public Health Medical Assistant can be described as a solution to the problem of the lack of access to reliable healthcare information, especially in the rural and semi-urban areas. The system has been modeled on a Retrieval-Augmented Generation (RAG) methodology coupled with semantic search based on a FAISS vector database to make sure that the answers are based on the proven medical knowledge. It was designed with a modular architecture which comprised of functional modules including symptom analysis, drug information retrieval, hospital finder and a medical knowledge engine. Moreover, a deterministic routing mechanism and safety layer was introduced in order to make responses context-aware, structured and user-safe.

The system implementation is a success that achieves the goals set in the introduction. The chatbot can comprehend queries of the users, give them the right healthcare information and direct the user to the right action including seeking medical assistance in case they need it. The inclusion of the WhatsApp platform makes it very accessible, and the user can communicate with the system using the communication platform that he or she is familiar with. The system

is also safe because it will prevent diagnosis and prescription and encourage awareness and preventive health care.

The analysis of the results shows that the system has about 80% accuracy in responding, as well as, high context relevance and clarity. RAG greatly suppresses hallucinated responses as opposed to the classical generative models. Moreover, the chatbot gives structured and understandable responses that are more user-friendly as compared to the traditional search engines, particularly to the non-technical individuals. The system also exhibits the good performance which has low response latency and consistent behavior of the system with the various query types.

In general, the project shows that AI-based conversational systems can be useful instruments of public health awareness through provision of correct, available, and safe information. The retrieval-based grounding and the modular design, as well as the real-time implementation, make the system applicable in the real-life scenario in healthcare communication.

9.1 Future Work

Despite the positive outcomes of the system, it is possible to improve it by making several changes to increase its capacity and usability. Among the most important improvements, the introduction of multilingual support should be mentioned, which will allow the chatbot to speak the local languages and attract a larger audience. This will be crucial in enhancing accessibility in rural regions where English might be a problem.

The other possible enhancement will be the use of SMS-based interaction whereby, the user and the internet with smart phones or constant internet connection are not connected, then the user can access the system. This would further increase the coverage of the chatbot to the underserved populations.

It is also possible to improve the system with the introduction of real-time public health databases that would automatically update the knowledge base with the latest medical information. Also, by adding predictive analytics and outbreak detection systems, the chatbot can be used to deliver proactive health notifications and preventive-oriented advice.

Additional enhancements can be optimization of the response time and improvement of the routing accuracy with the help of sophisticated NLP methods and implementation of the system on the scalable cloud infrastructure to serve a large population of users.

Such improvements in the future will help in making the system stronger, scaled and more effective in promoting the public health awareness and education.

References

[1] L. Laranjo et al., "Conversational agents in healthcare: A systematic review," *Journal of the American Medical Informatics Association*, vol. 25, no. 9, pp. 1248–1258, 2018.

[2] T. W. Bickmore et al., "Safety first: Conversational agents for health information," *Journal of Biomedical Informatics*, vol. 82, pp. 1–9, 2018.

[3] L. Wilson and M. Marasoiu, "The development and use of chatbots in public health: Scoping review," *JMIR Human Factors*, vol. 9, no. 4, p. e35882, 2022.

[4] F. Branda et al., "The role of AI-based chatbots in public health emergencies: A narrative review," *Future Internet*, vol. 17, no. 4, p. 145, 2025.

[5] K. Singhal et al., "Large language models encode clinical knowledge," *Nature*, vol. 620, pp. 172–180, 2023.

[6] Z. Ji et al., "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[7] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[8] G. Izacard and E. Grave, "Leveraging passage retrieval with generative models for open-domain question answering," in *Proc. EACL*, pp. 874–880, 2021.

[9] A. S. Miner et al., "Chatbots in the fight against the COVID-19 pandemic," *npj Digital Medicine*, vol. 3, no. 65, pp. 1–4, 2020.

[10] A. Devaraj et al., "Evidence-based clinical decision support using natural language processing," *Journal of Biomedical Informatics*, vol. 128, p. 104002, 2022.

[11] World Bank, "Rural population (% of total population) – India," *World Development Indicators*, World Bank, Washington, DC, USA.

[12] Ministry of Health and Family Welfare (MoHFW), *Rural Health Statistics 2022–23*, Government of India, New Delhi, India, 2023.

[13] World Health Organization (WHO), *Preventing Disease Through Healthy Environments: Global Health Estimates*, WHO Press, Geneva, Switzerland, 2020.

[14] Telecom Regulatory Authority of India (TRAI), *The Indian Telecom Services Performance Indicators Report*, TRAI, New Delhi, India, 2023.

[15] United Nations, *Transforming our world: The 2030 Agenda for Sustainable Development*, United Nations, New York, NY, USA, 2015.

[16] United Nations, "Sustainable Development Goal 3: Good Health and Well-Being," 2023.

[17] United Nations, "Sustainable Development Goal 10: Reduced Inequalities," 2023.

[18] United Nations, "Sustainable Development Goal 9: Industry, Innovation and Infrastructure," 2023.

[19] K. Beck et al., *Manifesto for Agile Software Development*, Agile Alliance, 2001.

[20] L. Floridi et al., "AI4People—An ethical framework for a good AI society," *Minds and Machines*, vol. 28, no. 4, pp. 689–707, 2018.

[21] Government of India, *Digital Personal Data Protection Act (DPDPA)*, 2023.

[22] European Union, *General Data Protection Regulation (GDPR)*, 2016.

[23] United Nations Environment Programme, *Sustainability and Digital Transformation*, 2022.

[24] S. Amershi et al., "Guidelines for human-AI interaction," in *Proc. CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2019.

BASE PAPER

[3] L. Wilson and M. Marasoiu, "The development and use of chatbots in public health: Scoping review," *JMIR Human Factors*, vol. 9, no. 4, p. e35882, 2022.

Appendix

Appendix A: Data Sheets

Since the proposed system is software-based, traditional hardware datasheets are not applicable. Instead, the following technical documentation and specifications were used:

- **FAISS Vector Database Documentation**
  Used for efficient similarity search and document retrieval.
- **FastAPI Framework Documentation**
  Provides backend API handling and asynchronous request processing.
- **Twilio WhatsApp API Documentation**
  Enables real-time communication between users and the chatbot.
- **Google Places API Documentation**
  Used for retrieving nearby hospital information based on location.
- **HuggingFace Embedding Models Documentation**
  Used for generating semantic embeddings for RAG pipeline.

Appendix B: Dataset Information

The system utilizes curated medical knowledge sources:

- Disease and symptom descriptions

Datasets were pre-processed and stored in the FAISS vector database for retrieval.

Appendix C: GitHub

https://github.com/Stu-Affan/Medical-Chatbot

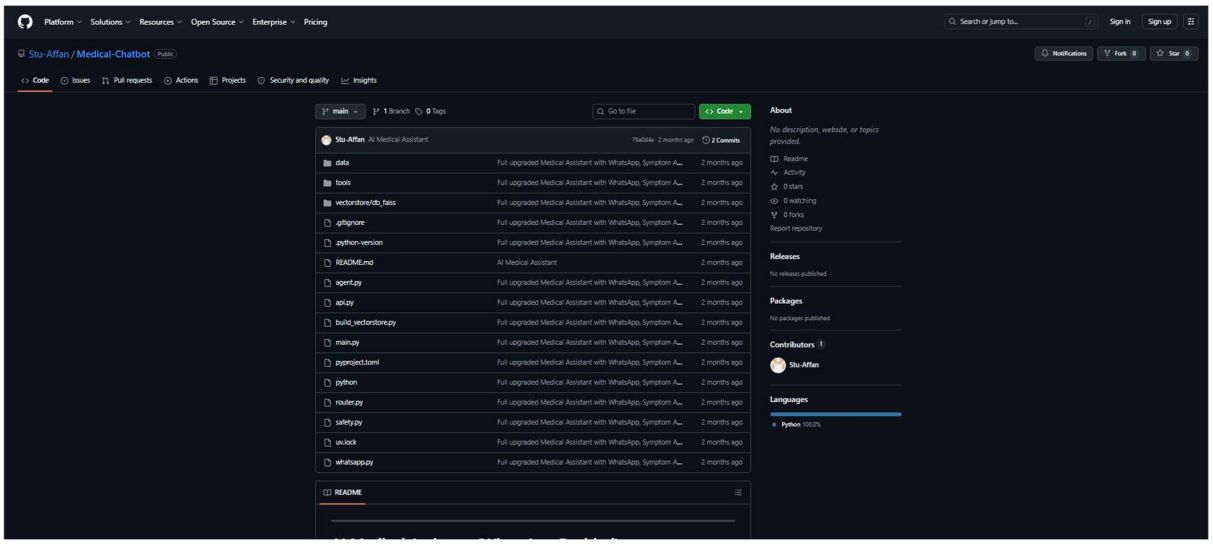


Fig. A.1 GitHub Repository

Appendix D: Project Images

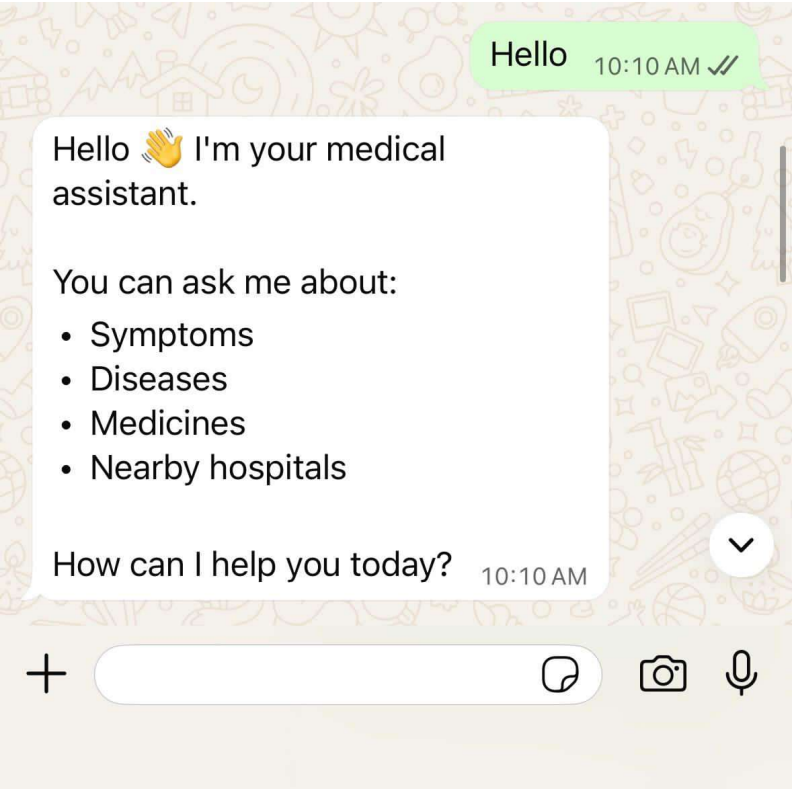
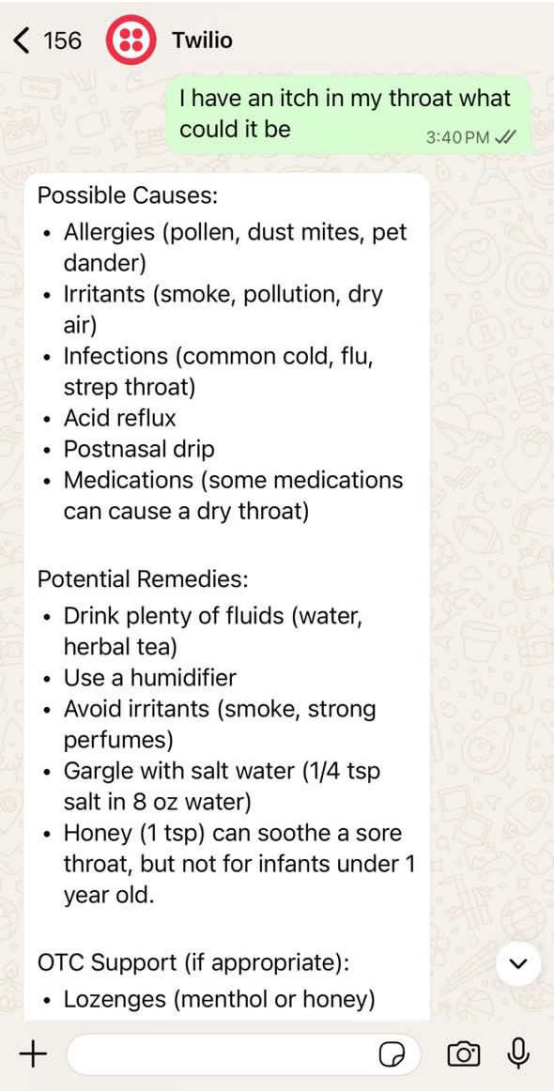


Fig. A.2 Chatbot type of desirable greetings and assisted queries over WhatsApp



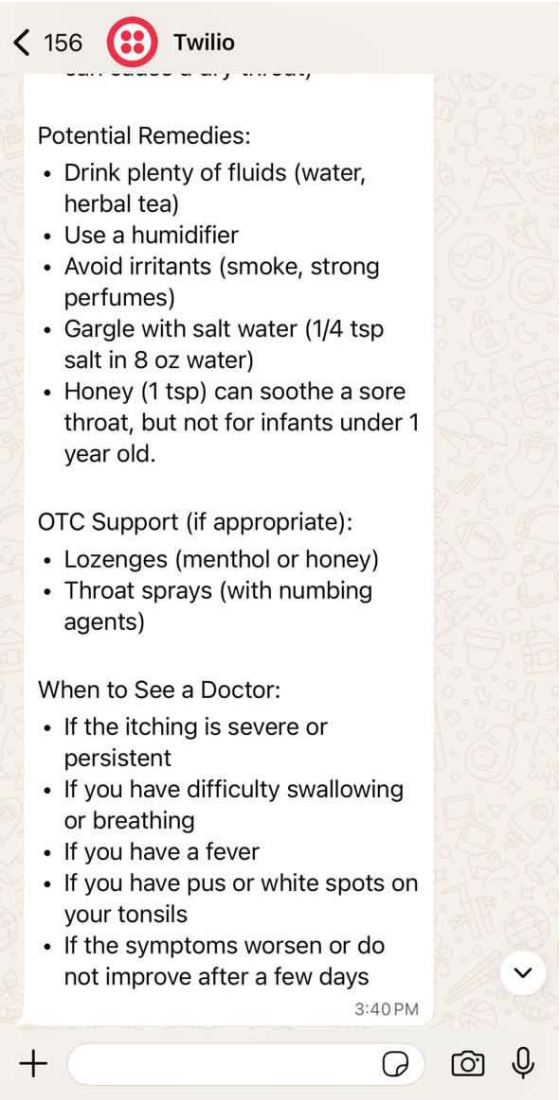
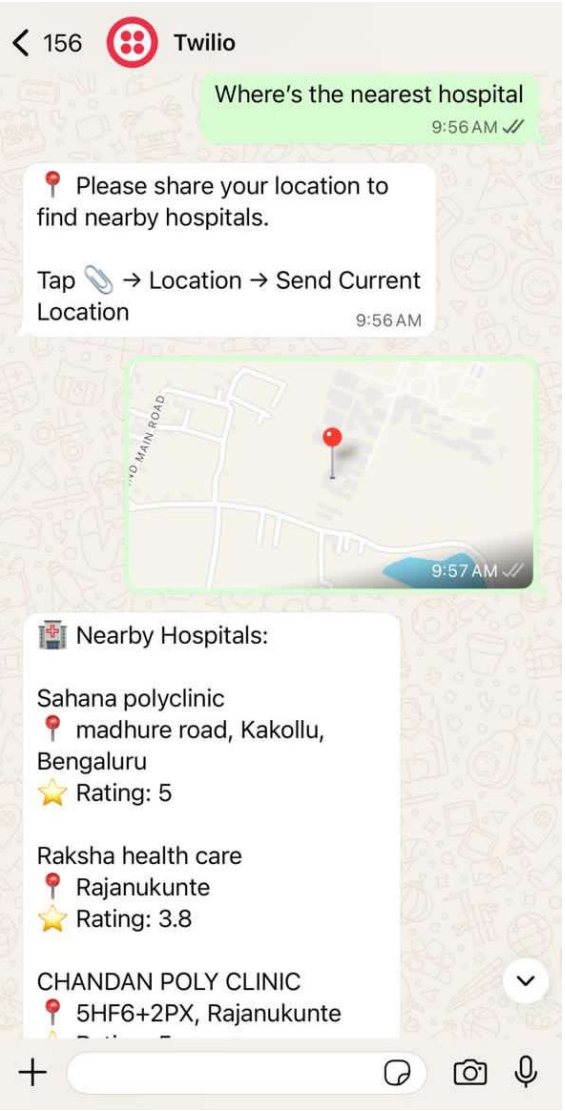


Fig A.3 Example of symptom analysis response generated by the chatbot



A.4 Nearby hospital discovery using location sharing through WhatsApp

Appendix E: Plagiarism Report

Appendix F: Publications